第三篇 多元函数微积分

第七章 多元函数微分学

§1 多元函数的极限与连续

习题 P.11.

1. 当 $(x,y)\to(0,0)$ 时，下列函数的极限是否存在？若存在，求出其极限.

(1) $\dfrac{x^3+xy^2}{x^2+y^2}$

解：当 $x^2+y^2\neq 0$ 时，$\dfrac{x^3+xy^2}{x^2+y^2}=x\to 0$. $((x,y)\to(0,0)$ 时$)$.

$\therefore \lim\limits_{(x,y)\to(0,0)}\dfrac{x^3+xy^2}{x^2+y^2}=0.$

(2) $\dfrac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}$

解：$\dfrac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}=\dfrac{(x^2+y^2)(\sqrt{x^2+y^2+1}+1)}{x^2+y^2+1-1}=\sqrt{x^2+y^2+1}+1\to 2$ $\left(\begin{matrix}x^2+y^2\neq 0\\(x,y)\to(0,0)\end{matrix}\right)$.

$\therefore \lim\limits_{(x,y)\to(0,0)}\dfrac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}=2.$

(3) $\dfrac{\sqrt{x^2y^2+1}-1}{x^2+y^2}$

解：$\dfrac{\sqrt{x^2y^2+1}-1}{x^2+y^2}=\dfrac{x^2y^2}{(x^2+y^2)(\sqrt{x^2y^2+1}+1)}\le\dfrac{x^2y^2}{2|xy|\cdot\sqrt{x^2y^2+1}}\le\dfrac{|xy|}{2\sqrt{x^2y^2+1}}$

$\le\dfrac{|xy|}{2\sqrt{2|xy|}}=\dfrac{1}{4}\sqrt{2|xy|}\le\dfrac{1}{4}\sqrt{x^2+y^2}.$

$\forall\varepsilon>0$，$\exists\delta=4\varepsilon$，$\forall(x,y)$，$0<\sqrt{x^2+y^2}<\delta$，成立：

$\left|\dfrac{\sqrt{x^2y^2+1}-1}{x^2+y^2}-0\right|\le\dfrac{1}{4}\sqrt{x^2+y^2}\le\dfrac{1}{4}\delta=\varepsilon.$ $\therefore\lim\limits_{(x,y)\to(0,0)}\dfrac{\sqrt{x^2y^2+1}-1}{x^2+y^2}=0.$

(4) $\dfrac{(x+y)^2}{x^2+y^2}$

解：$\because\lim\limits_{\substack{x\to 0\\y=-x}}\dfrac{(x+y)^2}{x^2+y^2}=0.$ $\lim\limits_{\substack{x\to 0\\y=x}}\dfrac{(x+y)^2}{x^2+y^2}=2.$ $\therefore$ 极限不存在.

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(5) $(x^2+y^2)e^{-(x+y)}$

解: $(0,0)$ 是初等函数. $\therefore \lim_{(x,y)\to(0,0)} (x^2+y^2)e^{-(x+y)} = 0$. (连续函数)

(6) $\dfrac{x^2-y^2}{x^2+y^2}$.

解: $\because \lim_{\substack{x\to 0\\ y=x}} \dfrac{x^2-y^2}{x^2+y^2} = 0$. $\lim_{\substack{x\to 0\\ y=2x}} \dfrac{x^2-y^2}{x^2+y^2} = \lim_{x\to 0} \dfrac{x^2-4x^2}{x^2+4x^2} = -\dfrac{3}{5}$. $\therefore$ 极限不存在.

(7) $\dfrac{1-\cos(x^2+y^2)}{x^2+y^2}$.

解: $\dfrac{1-\cos(x^2+y^2)}{x^2+y^2} \sim \dfrac{\frac{1}{2}(x^2+y^2)^2}{x^2+y^2} = \frac{1}{2}(x^2+y^2) \to 0$. $\left(\begin{matrix} x^2+y^2\neq 0,\\ (x,y)\to(0,0)\end{matrix}\right)$.

$\therefore \lim_{(x,y)\to(0,0)} \dfrac{1-\cos(x^2+y^2)}{x^2+y^2} = 0$.

(8) $\dfrac{x^2}{x^2+y^2-x}$.

解: $\because \lim_{\substack{x\to 0\\ y=x}} \dfrac{x^2}{x^2+y^2-x} = \lim_{x\to 0} \dfrac{x^2}{2x^2-x} = \lim_{x\to 0} \dfrac{x}{2x-1} = 0$.

$\lim_{\substack{x\to 0\\ y=\sqrt{x}}} \dfrac{x^2}{x^2+y^2-x} = \lim_{x\to 0} \dfrac{x^2}{x^2+x-x} = \lim_{x\to 0} 1 = 1$.

$\therefore$ 极限不存在.

2. 求下列极限:

a) $\lim_{(x,y)\to(1,0)} \dfrac{\ln(x+e^y)}{\sqrt{x^2+y^2}} = \dfrac{\ln(1+e^0)}{\sqrt{1+0}} = \dfrac{\ln 2}{1} = \ln 2$ (连续性)

b) $\lim_{(x,y)\to(0,0)} \left(\dfrac{1}{x^2}+\dfrac{1}{y^2}\right)e^{-\left(\frac{1}{x^2}+\frac{1}{y^2}\right)} = \lim_{(x,y)\to(0,0)} \left(\dfrac{1}{x^2}e^{-\frac{1}{x^2}}\right)\cdot e^{-\frac{1}{y^2}} + \lim_{(x,y)\to(0,0)} \left(\dfrac{1}{y^2}e^{-\frac{1}{y^2}}\right)\cdot e^{-\frac{1}{x^2}}$

$= 0\cdot 0 + 0\cdot 0 = 0$. (极限四则运算)

c) $\lim_{(x,y,z)\to(1,2,3)} \dfrac{xy^2-yz}{xyz-1} = \dfrac{1\times 4-4\times 3}{1\times 2\times 3-1} = -\dfrac{8}{5}$. (连续性)

(4) $\lim\limits_{(x,y,z)\to(0,0,0)} \dfrac{x^2y^2z^2}{x^2+y^2+z^2}$ (3阶无穷小量).

解: $x^2y^2z^2 \le (x^2+y^2+z^2)^3$ $\therefore \dfrac{x^2y^2z^2}{x^2+y^2+z^2} \le (x^2+y^2+z^2)^2 \to 0 \quad ((x,y,z)\to(0,0,0))$

$\therefore \lim\limits_{(x,y,z)\to(0,0,0)} \dfrac{x^2y^2z^2}{x^2+y^2+z^2} = 0.$

3. 讨论下列函数在原点 $O(0,0)$ 处是否连续?

(1) $z = \begin{cases} 1, & xy=0 \\ 0, & xy\neq 0 \end{cases}$

解: $f(x,y) = z = \begin{cases} 1 & xy=0 \\ 0 & xy\neq 0 \end{cases}$ $\because \lim\limits_{\substack{x\to 0 \\ y=x}} f(x,y) = 0$, $\lim\limits_{\substack{x\to 0 \\ y=0}} f(x,y) = 1$.

$\therefore f(x,y)$ 在 $(0,0)$ 处极限不存在. $\therefore$ 不连续.

(2) $z = \begin{cases} \dfrac{\sin(x^3+y^3)}{x^3+y^3}, & x^3+y^3\neq 0 \\ 0, & x^3+y^3=0 \end{cases}$

解: $\because \lim\limits_{\substack{x\to 0 \\ y=0}} f(x,y) = \lim\limits_{x\to 0} \dfrac{\sin x^3}{x^3} = 1 \neq 0 = f(0,0)$ $\therefore$ 不连续.

(3) $z = \begin{cases} \dfrac{\sin(x^3+y^3)}{x^2+y^2}, & x^2+y^2\neq 0 \\ 0, & x^2+y^2=0 \end{cases}$

解: $|\sin(x^3+y^3)| \le |x^3+y^3| \le |x+y|\cdot|x^2+y^2-xy| \le 2|x+y|\cdot|x^2+y^2| \le 2\sqrt{2}(x^2+y^2)^{\frac{3}{2}}$.

$\therefore \lim\limits_{(x,y)\to(0,0)} \dfrac{\sin(x^3+y^3)}{x^2+y^2} = 0 = f(0,0)$. $\therefore$ 连续.

4. 指出下列函数的连续范围.

(1) $u = \dfrac{1}{\sin x \cdot \sin y}$. $D = \{(x,y) \mid x \neq k\pi,\ y\neq l\pi,\ k,l \text{ 为整数}\}$.

(2) $u = \ln(1-x^2-y^2)$. $D = \{(x,y) \mid x^2+y^2 < 1\}$.

(3) $u = \ln\dfrac{1}{(x-a)^2+(y-b)^2}$. $D = \{(x,y) \mid (x,y)\neq(a,b)\}$.

5. 下列映射 $f: (x,y) \mapsto (u,v)$ 在 R^2 的哪个子集上连续？

(1) $\begin{cases} u = x^2 - y^2 \\ v = \dfrac{1}{x^2 - y^2} \end{cases}$ 解：在 $S = \{(x,y) \mid |x| \neq |y|,\ (x,y) \in R^2\}$ 上连续.

(2) $\begin{cases} u = \dfrac{x}{x^2+y^2} \\ v = \dfrac{y}{x^2+y^2} \end{cases}$ 解：在 $S = \{(x,y) \mid (x,y) \neq (0,0),\ (x,y) \in R^2\}$ 上连续.

§2. 全微分 §偏导数

习题. P.23.

1. 求下列函数的各个一阶偏导数：

(1) $z = x^3y + 3x^2y + xy^3$.

解：$\dfrac{\partial z}{\partial x} = 3x^2y + 6xy + y^3$. $\dfrac{\partial z}{\partial y} = x^3 + 3x^2 + 3xy^2$.

(2) $z = \dfrac{y}{x} + \dfrac{x}{y}$

解：$\dfrac{\partial z}{\partial x} = -\dfrac{y}{x^2} + \dfrac{1}{y}$. $\dfrac{\partial z}{\partial y} = \dfrac{1}{x} - \dfrac{x}{y^2}$.

(3) $z = \ln \tan \dfrac{y}{x}$.

解：$\dfrac{\partial z}{\partial x} = \dfrac{1}{\tan \frac{y}{x}} \cdot \sec^2 \dfrac{y}{x} \cdot \left(-\dfrac{y}{x^2}\right) = -\dfrac{y}{x^2 \sin \frac{y}{x} \cos \frac{y}{x}}$.

$\dfrac{\partial z}{\partial y} = \dfrac{1}{\tan \frac{y}{x}} \cdot \sec^2 \dfrac{y}{x} \cdot \dfrac{1}{x} = \dfrac{1}{x \sin \frac{y}{x} \cos \frac{y}{x}}$.

(4) $z = x^{xy}$

解：$\ln z = xy \ln x$

$\dfrac{\partial z}{\partial x} = z \cdot \left(y \ln x + xy \cdot \dfrac{1}{x}\right) = x^{xy} \cdot y (\ln z + 1)$.

$\dfrac{\partial z}{\partial y} = z (x \ln x) = x^{xy+1} \ln x$.

2. 计算下列函数在指定点的偏导数：

11) $z=\arcsin\frac{x}{y}$，求 $(1,2)$ 处的 z'_x, z'_y.

解：$z'_x=\frac{1}{\sqrt{1-\frac{x^2}{y^2}}}\cdot\frac{1}{y}=\frac{|y|}{\sqrt{y^2-x^2}}$. $z'_x(1,2)=\frac{1}{\sqrt{3}}$.

$z'_y=\frac{1}{\sqrt{1-\frac{x^2}{y^2}}}\left(-\frac{x}{y^2}\right)=-\frac{x}{|y|}\cdot\frac{1}{\sqrt{y^2-x^2}}$. $z'_y(1,2)=-\frac{1}{2\sqrt{3}}$.

12) $u=\frac{1}{\sqrt{x^2+y^2+z^2}}$，求 $(1,2,-1)$ 处的 u'_x, u'_y, u'_z.

解：$u'_x=-\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}}\cdot(2x)=-x(x^2+y^2+z^2)^{-\frac{3}{2}}$. $u'_x(1,2,-1)=-\frac{\sqrt{6}}{36}$.

$u'_y=-\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}}\cdot(2y)=-y(x^2+y^2+z^2)^{-\frac{3}{2}}$. $u'_y(1,2,-1)=-\frac{\sqrt{6}}{18}$.

$u'_z=-\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}}\cdot(2z)=-z(x^2+y^2+z^2)^{-\frac{3}{2}}$. $u'_z(1,2,-1)=\frac{\sqrt{6}}{36}$.

13) $u=e^{3x+4y}\cos(2x+5z)$ 求 $(-2,1,2)$ 处的 u'_x, u'_y, u'_z.

解：$u'_x=3e^{3x+4y}\cos(2x+5z)-2e^{3x+4y}\sin(2x+5z)=e^{3x+4y}[3\cos(2x+5z)-2\sin(2x+5z)]$

$u'_y=4e^{3x+4y}\cos(2x+5z)$. $u'_z=-5e^{3x+4y}\sin(2x+5z)$.

$\therefore u'_x(-2,1,2)=e^{-2}(3\cos b-2\sin b)$. $u'_y(-2,1,2)=4e^{-2}\cos b$.

$u'_z(-2,1,2)=-5e^{-2}\sin b$.

14) $u=\sin^2 x+\sin\left[(y-1)\ln\tan\sqrt{\frac{x}{y}}\right]$ 求 $(\frac{\pi}{4},1)$ 处的 u'_x

解：$u'_x=2\sin x\cos x+\cos\left[(y-1)\ln\tan\sqrt{\frac{x}{y}}\right]\cdot(y-1)\cdot\cot\sqrt{\frac{x}{y}}\cdot\sec^2\sqrt{\frac{x}{y}}\cdot 2\sqrt{\frac{y}{x}}\cdot\frac{1}{y}$

$u'_x(\frac{\pi}{4},1)=2\sin\frac{\pi}{4}\cos\frac{\pi}{4}=1$.

3. 求下列函数的全微分：

11) $z=ax^2y+bx^2$.

解：$\frac{\partial z}{\partial x}=2axy+2bx$. $\frac{\partial z}{\partial y}=ax^2$. $\therefore du=(2axy+2bx)dx+ax^2dy$.

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(2) $z = \tan^2(x^2+y^2)$

解: $\frac{\partial z}{\partial x} = 2\tan(x^2+y^2)\cdot\sec^2(x^2+y^2)\cdot 2x = 4x\tan(x^2+y^2)\cdot\sec^2(x^2+y^2)$.

$\frac{\partial z}{\partial y} = 4y\cdot\tan(x^2+y^2)\cdot\sec^2(x^2+y^2)$ ∴ $dz = 4\tan(x^2+y^2)\cdot\sec^2(x^2+y^2)\cdot(x\,dx+y\,dy)$.

(3) $z = \ln\left(x+\sqrt{x^2-y^2}\right)$

解: $\frac{\partial z}{\partial x} = \frac{1}{x+\sqrt{x^2-y^2}}\cdot\left(1+\frac{2x}{2\sqrt{x^2-y^2}}\right) = \frac{1}{\sqrt{x^2-y^2}}$.

$\frac{\partial z}{\partial y} = \frac{1}{x+\sqrt{x^2-y^2}}\cdot\frac{-2y}{2\sqrt{x^2-y^2}} = -\frac{y}{\sqrt{x^2-y^2}}\cdot\frac{1}{x+\sqrt{x^2-y^2}}$.

∴ $dz = \frac{dx}{\sqrt{x^2-y^2}} - \frac{y\,dy}{\sqrt{x^2-y^2}\cdot(x+\sqrt{x^2-y^2})}$.

(4) $z = xe^{-y} + ye^{-x}$

解: $\frac{\partial z}{\partial x} = e^{-y} - ye^{-x}$. $\frac{\partial z}{\partial y} = -xe^{-y} + e^{-x}$.

∴ $dz = (e^{-y} - ye^{-x})\,dx + (-xe^{-y} + e^{-x})\,dy$.

(5) $z = \arctan\frac{y}{x^2}$.

解: $\frac{\partial z}{\partial x} = \frac{1}{1+\frac{y^2}{x^4}}\cdot\frac{-2y}{x^3} = \frac{-2xy}{x^4+y^2}$. $\frac{\partial z}{\partial y} = \frac{1}{1+\frac{y^2}{x^4}}\cdot\frac{1}{x^2} = \frac{x^2}{x^4+y^2}$.

∴ $dz = \frac{-2xy\,dx + x^2dy}{x^4+y^2}$.

(6) $z = \int_x^y e^{t^2}dt$

解: $\frac{\partial z}{\partial x} = -e^{x^2}$ $\frac{\partial z}{\partial y} = e^{y^2}$. ∴ $dz = -e^{x^2}dx + e^{y^2}dy$.

4. 求曲线 $\begin{cases} z = \frac{1}{4}(x^2+y^2) \\ y = 2 \end{cases}$ 在点 $M_0(4,2,5)$ 处的切线关于x轴的倾角，并求该切线的方程.

解: $\frac{\partial z}{\partial x} = \frac{x}{2}$. 在 $M_0(4,2,5)$ 处: $\tan\theta = \left.\frac{\partial z}{\partial x}\right|_{x_0=4} = 2$. ∴ $\theta = \arctan 2$.

切线方程: $\begin{cases} z-5 = 2(x-4) \\ y = 2 \end{cases}$ 即 $\begin{cases} 2x-z-3=0 \\ y=2 \end{cases}$.

5. 讨论下列函数在 $O(0,0)$ 处的可微性.

(1) $z=\sqrt{x}\cos y$

解: $\dfrac{\partial z}{\partial x}=\lim\limits_{\Delta x\to 0}\dfrac{\sqrt{x+\Delta x}\cos y-\sqrt{x}\cdot\cos y}{\Delta x}=\lim\limits_{\Delta x\to 0}\dfrac{\cos y}{\sqrt{x+\Delta x}+\sqrt{x}}$.

$\dfrac{\partial z}{\partial x}(0,0)=\lim\limits_{\Delta x\to 0}\dfrac{1}{\sqrt{\Delta x}+0}=\lim\limits_{\Delta x\to 0}\dfrac{1}{\sqrt{\Delta x}}$. 极限不存在. $\therefore$ z 在 $O(0,0)$ 不可微.

(2) $z=\begin{cases}\dfrac{2xy}{\sqrt{x^2+y^2}} & x^2+y^2\neq 0.\\ 0 & x^2+y^2=0.\end{cases}$

解: $\Delta z=f(x+\Delta x,y+\Delta y)-f(x,y)=f(\Delta x,\Delta y)-f(0,0)=\dfrac{2\Delta x\Delta y}{\sqrt{\Delta x^2+\Delta y^2}}$.

(函数 $z=f(x,y)$ 在 $(0,0)$ 连续, 在 $(0,0)$ 的邻域内 f'_x 和 f'_y 有界. 但在 $O(0,0)$ 不可微)

$\because$ $|f(x,y)-f(0,0)|=\left|\dfrac{2xy}{\sqrt{x^2+y^2}}\right|\leq\left|\dfrac{x^2+y^2}{\sqrt{x^2+y^2}}\right|=\sqrt{x^2+y^2}$.

$\forall\varepsilon>0$. $\exists\delta=\varepsilon$. $\forall 0\leq\sqrt{x^2+y^2}<\delta$. 成立: $|f(x,y)-f(0,0)|<\varepsilon$.

$\therefore$ $f(x,y)$ 在 $O(0,0)$ 连续.

当 $x^2+y^2\neq 0$ 时, $f'_x(x,y)=\left(\dfrac{y^2}{x^2+y^2}\right)^{3/2}$. $f'_y(x,y)=\left(\dfrac{x^2}{x^2+y^2}\right)^{3/2}$.

在 $(0,0)$ 的邻域内 f'_x 和 f'_y 都为初等连续函数. 有界.

当 $x^2+y^2=0$ 时. 即 $O(0,0)$ 处:

$f'_x(0,0)=\lim\limits_{\Delta x\to 0}\dfrac{f(0+\Delta x,0)-f(0,0)}{\Delta x}=\lim\limits_{\Delta x\to 0}\dfrac{f(\Delta x,0)-f(0,0)}{\Delta x}=0$.

$f'_y(0,0)=\lim\limits_{\Delta y\to 0}\dfrac{f(0,0+\Delta y)-f(0,0)}{\Delta y}=\lim\limits_{\Delta y\to 0}\dfrac{f(0,\Delta y)-f(0,0)}{\Delta y}=0$.

$\therefore$ $f(x,y)$ 在 $O(0,0)$ 处. f'_x 和 f'_y 均存在.

但是在 $(0,0)$ 处: $\dfrac{\Delta z-f'_x(0,0)\cdot\Delta x-f'_y(0,0)\cdot\Delta y}{\sqrt{\Delta x^2+\Delta y^2}}=\dfrac{2\Delta x\Delta y}{\Delta x^2+\Delta y^2}\not\to 0$. $\left(\begin{matrix}\Delta x\to 0\\ \Delta y=\Delta x\end{matrix}\right\}\to 1$ $\left.\begin{matrix}\Delta x\to 0\\ \Delta y=0\end{matrix}\right\}\to 0$

即 $\Delta z\neq f'_x(0,0)\cdot\Delta x+f'_y(0,0)\cdot\Delta y+o(\sqrt{\Delta x^2+\Delta y^2})$.

$\therefore$ $z(x,y)$ 在 $O(0,0)$ 不可微.

6. 用全微分求下列函数在指定点的近似值：

(1) $\sqrt{20-x^2-7y^2}$ $(1.95, 1.08)$. (2.8342)

解: $f(x,y)=\sqrt{20-x^2-7y^2}$. $\frac{\partial f}{\partial x}=\frac{-x}{\sqrt{20-x^2-7y^2}}$ $\frac{\partial f}{\partial y}=\frac{-14y}{2\sqrt{20-x^2-7y^2}}$.

$f(1.95,1.08)\approx \frac{\partial f}{\partial x}(2,1)\cdot\Delta x+\frac{\partial f}{\partial y}(2,1)\cdot\Delta y+f(2,1)$

$=-\frac{2}{3}\cdot(-0.05)-\frac{7}{3}\cdot(0.08)+3=2.843$.

(2) $\ln(x-3y)$. $(6.9, 2.06)$. (-0.3285)

解: $f(x,y)=\ln(x-3y)$. $\frac{\partial f}{\partial x}=\frac{1}{x-3y}$. $\frac{\partial f}{\partial y}=\frac{-3}{x-3y}$.

$f(6.9,2.06)\approx \frac{\partial f}{\partial x}(7,2)\cdot\Delta x+\frac{\partial f}{\partial y}(7,2)\cdot\Delta y+f(7,2)$

$=1\times(-0.1)-3\times(0.06)+0=-0.28$.

7. 测得一矩形的长和宽分别为 20 cm 和 12 cm，可能的最大测量误差为 0.1 cm. 试用全微分估计由测量值计算此矩形面积的最大误差.

解: $S(x,y)=xy$. $\frac{\partial S}{\partial x}=y$. $\frac{\partial S}{\partial y}=x$. $dS=\frac{\partial S}{\partial x}dx+\frac{\partial S}{\partial y}dy=y\,dx+x\,dy$.

$\therefore dS\le y\cdot\delta_c+x\cdot\delta_c=12\times0.1+20\times0.1=3.2\ cm^2$.

$\therefore$ 面积的最大误差为 $3.2\ cm^2$.

8. 求下列函数的二阶偏导数 u''_{xx}. u''_{xy}. u''_{yy}.

(1) $u=\sin(ax-by)$

解: $u'_x=a\cos(ax-by)$. $u'_y=-b\cos(ax-by)$.

$u''_{xx}=-a^2\sin(ax-by)$. $u''_{xy}=ab\sin(ax-by)$. $u''_{yy}=-b^2\sin(ax-by)$.

(2) $u=e^{ax}\cos by$

解: $u'_x=ae^{ax}\cos by$. $u'_y=-be^{ax}\sin by$

$u''_{xx}=a^2e^{ax}\cos by$. $u''_{xy}=-abe^{ax}\sin by$. $u''_{yy}=-b^2e^{ax}\cos by$.

(3) $u=ye^{xy}$

解: $u'_x=y^2e^{xy}$. $u'_y=e^{xy}+xye^{xy}=(1+xy)e^{xy}$.

$u''_{xx}=y^3e^{xy}$. $u''_{xy}=2ye^{xy}+xy^2e^{xy}=(2+xy)ye^{xy}$ $u''_{yy}=x(2+xy)e^{xy}$.

(4) $u = x^{\ln y}$

解: $u'_x = \ln y \cdot x^{\ln y - 1}$ $\quad u'_y = \ln x \cdot x^{\ln y} \cdot \frac{1}{y}$.

$u''_{xx} = \ln y \cdot (\ln y - 1) \cdot x^{\ln y - 2} = \frac{1}{x^2} \ln y \cdot (\ln y - 1) \cdot x^{\ln y}$

$u''_{xy} = \frac{1}{xy} x^{\ln y} + \ln x \cdot \ln y \cdot x^{\ln y - 1} \cdot \frac{1}{y} = \frac{1}{xy} (\ln x \cdot \ln y + 1) \cdot x^{\ln y}$

$u''_{yy} = \ln x \cdot (-\frac{1}{y^2}) \cdot x^{\ln y} + \ln x \cdot \frac{1}{y} \cdot \ln x \cdot x^{\ln y} \cdot \frac{1}{y} = \frac{1}{y^2} (\ln^2 x - \ln x) \cdot x^{\ln y}$.

9. 设函数 $f(x,y) = \begin{cases} xy \frac{x^2-y^2}{x^2+y^2} & x^2+y^2 \neq 0 \\ 0 & x^2+y^2 = 0 \end{cases}$ 试求 $f'_x(0,y)$ 及 $f'_y(x,0)$, 并证明 $f''_{xy}(0,0) \neq f''_{yx}(0,0)$.

解: $f'_x(0,y) = \lim_{\Delta x \to 0} \frac{f(\Delta x, y) - f(0,y)}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta x \cdot y \cdot \frac{\Delta x^2 - y^2}{\Delta x^2 + y^2} - 0}{\Delta x} = -y$.

$\therefore f''_{xy}(0,0) = -1$.

$f'_y(x,0) = \lim_{\Delta y \to 0} \frac{f(x, \Delta y) - f(x,0)}{\Delta y} = \lim_{\Delta y \to 0} \frac{x \cdot \Delta y \cdot \frac{x^2 - \Delta y^2}{x^2 + \Delta y^2}}{\Delta y} = x$.

$\therefore f''_{yx}(0,0) = 1$.

$\therefore f''_{xy}(0,0) \neq f''_{yx}(0,0)$.

10. 设 $u = 2\cos^2(x - \frac{t}{2})$. 证明: $2\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x \partial t} = 0$.

证明: $\frac{\partial u}{\partial t} = 2 \cdot 2\cos(x - \frac{t}{2}) \cdot [-\sin(x - \frac{t}{2})] \cdot (-\frac{1}{2}) = 2\sin(x - \frac{t}{2})\cos(x - \frac{t}{2}) = \sin(2x - t)$.

$\frac{\partial^2 u}{\partial t^2} = -\cos(2x - t)$. $\quad \frac{\partial^2 u}{\partial x \partial t} = 2\cos(2x - t)$.

$\therefore 2\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x \partial t} = 0$.

11. 证明: 函数 $u(x,t) = \frac{1}{2a\sqrt{\pi t}} \cdot e^{-\frac{(x-b)^2}{4a^2 t}}$ 满足热传导方程 $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$.

证明: $\frac{\partial u}{\partial t} = \frac{1}{2a\sqrt{\pi}} \cdot (-\frac{1}{2}) \cdot \frac{1}{t\sqrt{t}} e^{-\frac{(x-b)^2}{4a^2 t}} + \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-b)^2}{4a^2 t}} \cdot \left(\frac{(x-b)^2}{4a^2} \cdot \frac{1}{t^2}\right)$

$= \frac{1}{2a\sqrt{\pi t}} e^{-\frac{(x-b)^2}{4a^2 t}} \cdot \left(-\frac{1}{2t} + \frac{(x-b)^2}{4a^2 t^2}\right)$

$\frac{\partial u}{\partial x} = \frac{1}{2a\sqrt{\pi t}} \cdot e^{-\frac{(x-b)^2}{4a^2 t}} \cdot \left(-\frac{2(x-b)}{4a^2 t}\right)$.

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{2a\sqrt{\pi t}} \cdot e^{-\frac{(x-b)^2}{4a^2t}} \cdot \left[\frac{4(x-b)^2}{(4a^2t)^2} - \frac{2}{4a^2t} \right] = \frac{1}{2a^3\sqrt{\pi t}} \cdot e^{-\frac{(x-b)^2}{4a^2t}} \left(-\frac{1}{2t} + \frac{(x-b)^2}{4a^2t^2} \right)$$

$$\therefore \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}.$$

12. 证明: n元函数 $u = (x_1^2 + x_2^2 + \cdots + x_n^2)^{\frac{2-n}{2}}$ 满足方程

$$u''_{x_1x_1} + u''_{x_2x_2} + \cdots + u''_{x_nx_n} = 0 \quad (n>2).$$

证明: $u'_{x_i} = \frac{2-n}{2}(x_1^2 + x_2^2 + \cdots + x_n^2)^{\frac{2-n}{2}-1} \cdot 2x_i = (2-n)x_i \cdot (x_1^2 + x_2^2 + \cdots + x_n^2)^{-\frac{n}{2}}$.

$$u''_{x_ix_i} = (2-n)\cdot(x_1^2 + x_2^2 + \cdots + x_n^2)^{-\frac{n}{2}} + (2-n)\cdot x_i \cdot \left(-\frac{n}{2}\right)\cdot(x_1^2 + x_2^2 + \cdots + x_n^2)^{-\frac{n}{2}-1} \cdot 2x_i$$

$$= (2-n)\cdot(x_1^2 + x_2^2 + \cdots + x_n^2)^{-\frac{n}{2}} \cdot \left(1 - \frac{n x_i^2}{x_1^2 + x_2^2 + \cdots + x_n^2}\right)$$

$$\therefore u''_{x_1x_1} + u''_{x_2x_2} + \cdots + u''_{x_nx_n} = (2-n)\cdot(x_1^2 + x_2^2 + \cdots + x_n^2)^{-\frac{n}{2}} \cdot \left(n - n\cdot\frac{x_1^2 + \cdots + x_n^2}{x_1^2 + \cdots + x_n^2}\right)$$

$$= 0.$$

13. 设映射 f 为 $(x,y)^T \mapsto (u,v)^T$, 其中 u,v 关系由下列函数组定义, 试求 f 的 Jacobi 矩阵及微分:

(1) $\begin{cases} u = e^x \cos y \\ v = e^x \sin y \end{cases}$

解:
$$J = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} e^x\cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{bmatrix} = e^x \begin{bmatrix} \cos y & -\sin y \\ \sin y & \cos y \end{bmatrix}$$

$$d\vec{f} = J \cdot d\vec{x} = J \cdot \begin{pmatrix} dx \\ dy \end{pmatrix} = e^x \begin{bmatrix} \cos y & -\sin y \\ \sin y & \cos y \end{bmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} = e^x \begin{pmatrix} \cos y\, dx - \sin y\, dy \\ \sin y\, dx + \cos y\, dy \end{pmatrix}$$

(2) $\begin{cases} u = \ln\sqrt{x^2+y^2} \\ v = \arctan\frac{y}{x} \end{cases}$

解:
$$J = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{x}{x^2+y^2} & \frac{y}{x^2+y^2} \\ -\frac{y}{x^2+y^2} & \frac{x}{x^2+y^2} \end{bmatrix} = \frac{1}{x^2+y^2}\begin{bmatrix} x & y \\ -y & x \end{bmatrix}.$$

$$d\vec{f} = \begin{pmatrix} du \\ dv \end{pmatrix} = J \cdot d\vec{x} = J \begin{pmatrix} dx \\ dy \end{pmatrix} = \frac{1}{x^2+y^2}\begin{bmatrix} x & y \\ -y & x \end{bmatrix}\begin{pmatrix} dx \\ dy \end{pmatrix} = \frac{1}{x^2+y^2}\begin{pmatrix} x\,dx + y\,dy \\ -y\,dx + x\,dy \end{pmatrix}$$

14. 计算下列映射的导数：

(1) $\vec{f}(x,y)=\begin{pmatrix} x+y \\ x^2+y^2 \end{pmatrix}$.

解：$\vec{f}(x,y)=\begin{pmatrix} f_1(x,y) \\ f_2(x,y) \end{pmatrix}=\begin{pmatrix} x+y \\ x^2+y^2 \end{pmatrix}$.

$$\vec{f}'(x,y)=J=\begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}=\begin{bmatrix} 1 & 1 \\ 2x & 2y \end{bmatrix}.$$

(2) $\vec{g}(u,v)=\begin{bmatrix} u\cos v \\ u\sin v \\ v \end{bmatrix}$

解：$\vec{g}(u,v)=\begin{pmatrix} g_1(u,v) \\ g_2(u,v) \\ g_3(u,v) \end{pmatrix}=\begin{pmatrix} u\cos v \\ u\sin v \\ v \end{pmatrix}$

$$\vec{g}'(u,v)=J=\begin{bmatrix} \frac{\partial g_1}{\partial u} & \frac{\partial g_1}{\partial v} \\ \frac{\partial g_2}{\partial u} & \frac{\partial g_2}{\partial v} \\ \frac{\partial g_3}{\partial u} & \frac{\partial g_3}{\partial v} \end{bmatrix}=\begin{bmatrix} \cos v & -u\sin v \\ \sin v & u\cos v \\ 0 & 1 \end{bmatrix}$$

15. 求曲面 $z=2x^2+4y^2$ 在 $(2,1,12)$ 处的切平面方程.

解：$$\vec{n}=\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & f'_x(x_0,y_0) \\ 0 & 1 & f'_y(x_0,y_0) \end{vmatrix}=\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 4x_0 \\ 0 & 1 & 8y_0 \end{vmatrix}=-4x_0\vec{i}-8y_0\vec{j}+\vec{k}.$$

切平面方程：$4x_0\cdot(x-x_0)+8y_0\cdot(y-y_0)-(z-z_0)=0$. $(x_0,y_0,z_0)=(2,1,12)$.

即：$8(x-2)+8(y-1)-(z-12)=0 \Rightarrow 8x+8y-z-12=0$.

16. 求螺旋线 $\vec{r}(t)=2\cos t\cdot\vec{i}+\sin t\cdot\vec{j}+t\vec{k}$ 在点 $(0,1,\frac{\pi}{2})$ 处的切线方程.

解：切向量 $\vec{r}'(t)=(-2\sin t,\cos t,1)$.

在点 $(0,1,\frac{\pi}{2})$ 处，$t=\frac{\pi}{2}$. $\vec{r}'(t)=(-2,0,1)$.

∴ 切线方程：$\frac{x}{-2}=\frac{y-1}{0}=\frac{z-\frac{\pi}{2}}{1}$ 即：$\begin{cases} y=1 \\ x+2z=\pi \end{cases}$

17. 求曲线 $x=t-\sin t$, $y=1-\cos t$, $z=4\sin\frac{t}{2}$ 在点 $(\frac{\pi}{2}-1,1,2\sqrt{2})$ 处的切线方程.

解：t 的向量 $\vec{r}'(t)=(1-\cos t,\ \sin t,\ 2\cos\frac{t}{2})$.

将点 $(\frac{\pi}{2}-1,\ 1,\ 2\sqrt{2})$ 处：$t=\frac{\pi}{2}$，$\vec{r}'(t)=(1,\ 1,\ \sqrt{2})$.

∴ 切线方程：$x-\frac{\pi}{2}+1=y-1=\frac{z-2\sqrt{2}}{\sqrt{2}}$.

18. 求曲线 $x=t$，$y=t^2$，$z=t^3$ 上的切线平行于平面 $x+2y+z=4$ 的点.

解：平面法向量 $\vec{n}=(1,\ 2,\ 1)$. 曲线上的向量 $\vec{r}'(t)=(1,\ 2t,\ 3t^2)$.

由 $\vec{n}\cdot\vec{r}'(t)=0 \Rightarrow 1+4t+3t^2=0 \Rightarrow t=-1$ 或 $t=-\frac{1}{3}$.

∴ 所求点为 $(-1,\ 1,\ -1)$ 或 $(-\frac{1}{3},\ \frac{1}{9},\ -\frac{1}{27})$.

§3. 链式求导法则

习题 · P34.

1. 设 $z=u^2\ln v$，其中 $u=\frac{x}{y}$，$v=3x-2y$，求 $\frac{\partial z}{\partial x}$，$\frac{\partial z}{\partial y}$.

解：$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}=2u\cdot\ln v\cdot\frac{1}{y}+\frac{u^2}{v}\cdot 3=\frac{2x}{y^2}\ln(3x-2y)+\frac{3x^2}{y^2(3x-2y)}$.

$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}=2u\cdot\ln v\cdot(-\frac{x}{y^2})+\frac{u^2}{v}\cdot(-2)=-\frac{2x^2}{y^3}\ln(3x-2y)-\frac{2x^2}{y^2(3x-2y)}$.

2. 设 $w=x^2+y^2+\sin(x+y)$，$x=u+v$，$y=uv$，求 $\frac{\partial w}{\partial u}$，$\frac{\partial w}{\partial v}$.

解：$\frac{\partial w}{\partial u}=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial u}=[2x+\cos(x+y)]\cdot 1+[2y+\cos(x+y)]\cdot v$

$=2(u+v)+\cos(u+v+uv)+2uv^2+v\cos(u+v+uv)$

$=2(u+v+uv^2)+(1+v)\cos(u+v+uv)$.

$\frac{\partial w}{\partial v}=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial v}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial v}=[2x+\cos(x+y)]\cdot 1+[2y+\cos(x+y)]\cdot u$

$=2(u+v)+\cos(u+v+uv)+2u^2v+u\cos(u+v+uv)$

$=2(u+v+u^2v)+(1+u)\cos(u+v+uv)$.

3. 设 $z=x^2y-xy^2$，$x=r\cos\theta$，$y=r\sin\theta$，求 $\frac{\partial z}{\partial r}$，$\frac{\partial z}{\partial\theta}$.

解：$\frac{\partial z}{\partial r}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial r}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial r}=(2xy-y^2)\cdot\cos\theta+(x^2-2xy)\cdot\sin\theta$

$= r\sin\theta\cos\theta(2r\cos\theta - r\sin\theta) + r\sin\theta\cos\theta(r\cos\theta - 2r\sin\theta)$

$= r\sin\theta\cos\theta(3r\cos\theta - 3r\sin\theta) = 3r^2\sin\theta\cos\theta(\cos\theta - \sin\theta)$.

$\frac{\partial z}{\partial\theta} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial\theta} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial\theta} = (2xy - y^2)\cdot(-r\sin\theta) + (x^2 - 2xy)\cdot(r\cos\theta)$

$= -r^2\sin^2\theta\cdot(2r\cos\theta - r\sin\theta) + r^2\cos^2\theta\cdot(r\cos\theta - 2r\sin\theta)$

$= r^3(\sin^3\theta - 2\sin^2\theta\cos\theta - 2\sin\theta\cos^2\theta + \cos^3\theta) = r^3(\sin\theta+\cos\theta)(1-3\sin\theta\cos\theta)$.

4. 设 $z = \arccos(x+y)$, $x = 3t$, $y = 4t^3$. 求 $\frac{dz}{dt}$.

解: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial t} = -\frac{1}{\sqrt{1-(x+y)^2}}\cdot 3 - \frac{1}{\sqrt{1-(x+y)^2}}\cdot 12t^2$

$= -\frac{3+12t^2}{\sqrt{1-9t^2-24t^4-16t^6}}$.

5. 设 $z = \arctan(xy)$. 而 ~~$y = a\sin x$~~ $y = e^x$. 求 $\frac{dz}{dx}$.

解: $\frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial x} = \frac{1}{1+x^2y^2}\cdot y + \frac{1}{1+x^2y^2}\cdot x\cdot e^x = \frac{(1+x)e^x}{1+x^2e^{2x}}$.

6. 设 $u = \frac{e^{ax}(y-z)}{a^2+1}$, $y = a\sin x$, $z = \cos x$. 求 $\frac{du}{dx}$.

解: $\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial x} + \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial x} = \frac{a(y-z)e^{ax}}{a^2+1} + \frac{e^{ax}}{a^2+1}\cdot a\cos x - \frac{e^{ax}}{a^2+1}(-\sin x)$

$= \frac{e^{ax}}{a^2+1}\left[a(a\sin x - \cos x) + a\cos x + \sin x\right] = e^{ax}\sin x$.

7. 设 f 具有连续一阶偏导数, 求 $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$. 其中:

(1) $u = f(x^2-y^2, e^{xy})$

解: $u = f(w, v)$, $w = x^2-y^2$, $v = e^{xy}$.

$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial w}\cdot\frac{\partial w}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x} = f_1'\cdot 2x + f_2'\cdot y e^{xy} = 2x\cdot f_1' + ye^{xy}\cdot f_2'$.

$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial w}\cdot\frac{\partial w}{\partial y} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y} = f_1'\cdot(-2y) + f_2'\cdot x e^{xy} = -2y\cdot f_1' + x e^{xy}\cdot f_2'$.

(2) $u = f\left(\frac{x}{y}, \frac{y}{x}\right)$.

解: $u = f(w, v)$. $w = \frac{x}{y}$, $v = \frac{y}{x}$.

$$\frac{\partial u}{\partial x}=\frac{\partial f}{\partial w}\cdot\frac{\partial w}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}=\frac{1}{y}f_1'+f_2'\cdot\left(-\frac{y}{x^2}\right)=\frac{1}{y}f_1'-\frac{y}{x^2}f_2'.$$

$$\frac{\partial u}{\partial y}=\frac{\partial f}{\partial w}\cdot\frac{\partial w}{\partial y}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y}=f_1'\cdot\left(-\frac{x}{y^2}\right)+f_2'\cdot\left(\frac{1}{x}\right)=-\frac{x}{y^2}f_1'+\frac{1}{x}f_2'.$$

8. 设 $f(x,y)$ 具有连续偏导数，且 $f(1,1)=1$，$f_x'(1,1)=2$，$f_y'(1,1)=3$，又设

$\varphi(x)=f(x,f(x,x))$，求 $\varphi'(1)$.

解：$\varphi(x)=f(x,y)$，$y=f(x,x)$.

$$\frac{dy}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cdot\qquad\varphi'(x)=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cdot\frac{dy}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\left(\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\right)$$

$$=f_x'+f_y'(f_x'+f_y')$$

$$\therefore\ \varphi'(1)=f_x'(1,1)+f_y'(1,1)\cdot\left[f_x'(1,1)+f_y'(1,1)\right]$$

$$=2+3\cdot(2+3)=17.$$

9. 设 f 是可微函数，a,b 为常数，$z=f(x+at,y+bt)$. 证明：

$$\frac{\partial z}{\partial t}=a\frac{\partial z}{\partial x}+b\frac{\partial z}{\partial y}.$$

证明：$\frac{\partial z}{\partial t}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial t}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial t}$. 其中：$\begin{cases}u=x+at\\ v=y+bt\end{cases}$ $\frac{\partial u}{\partial t}=a$，$\frac{\partial u}{\partial t}=b$.

而：$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}=\frac{\partial f}{\partial u}$.

$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y}=\frac{\partial f}{\partial v}$.

$\therefore\ \frac{\partial z}{\partial t}=a\frac{\partial z}{\partial x}+b\frac{\partial z}{\partial y}$.

10. 设 f 是可微函数，$u=xy+xf\left(\frac{y}{x}\right)$. 证明：$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=u+xy$.

证明：$\frac{\partial u}{\partial x}=y+f\left(\frac{y}{x}\right)+x\cdot f'\left(\frac{y}{x}\right)\cdot\left(-\frac{y}{x^2}\right)=y+f\left(\frac{y}{x}\right)-\frac{y}{x}f'\left(\frac{y}{x}\right)$.

$\frac{\partial u}{\partial y}=x+x\cdot f'\left(\frac{y}{x}\right)\cdot\frac{1}{x}=x+f'\left(\frac{y}{x}\right)$.

$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=xy+xf\left(\frac{y}{x}\right)-yf'\left(\frac{y}{x}\right)+xy+yf'\left(\frac{y}{x}\right)$

$=xy+xf\left(\frac{y}{x}\right)+xy=u+xy$.

11. 设 f 是二元函数，具有二阶连续偏导数，求下列函数的 $\frac{\partial^2 z}{\partial x^2}$、$\frac{\partial^2 z}{\partial x \partial y}$、$\frac{\partial^2 z}{\partial y^2}$。

(1) $z = f(xy, y)$。

解：$\frac{\partial z}{\partial x} = f_1' \cdot y = y f_1'$。 $\frac{\partial z}{\partial y} = f_1' \cdot x + f_2' = x f_1' + f_2'$。

$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}(y f_1') = y(f_{11}'' \cdot y) = y^2 f_{11}''$。

$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}(y f_1') = f_1' + y \cdot \frac{\partial}{\partial y}(f_1') = f_1' + y \cdot (f_{11}'' \cdot x + f_{12}'')$

$= f_1' + xy f_{11}'' + y f_{12}''$。

$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}(x f_1' + f_2') = x(f_{11}'' \cdot x + f_{12}'') + f_{21}'' \cdot x + f_{22}' = x^2 f_{11}'' + 2x f_{12}'' + f_{22}''$。

(2) $z = f\left(x, \frac{x}{y}\right)$。

解：$\frac{\partial z}{\partial x} = f_1' \cdot 1 + f_2' \cdot \frac{1}{y} = f_1' + \frac{1}{y} f_2'$。 $\frac{\partial z}{\partial y} = f_1' \cdot 0 + f_2' \cdot \left(-\frac{x}{y^2}\right) = -\frac{x}{y^2} f_2'$。

$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(f_1' + \frac{1}{y} f_2'\right) = f_{11}'' \cdot 1 + f_{12}'' \cdot \frac{1}{y} + \frac{1}{y} f_{21}'' \cdot 1 + \frac{1}{y} f_{22}'' \cdot \frac{1}{y} = f_{11}'' + \frac{2}{y} f_{12}'' + \frac{1}{y^2} f_{22}''$。

$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}\left(-\frac{x}{y^2} f_2'\right) = -\frac{1}{y^2} f_2' - \frac{x}{y^2}\left(f_{21}'' \cdot 1 + f_{22}'' \cdot \frac{1}{y}\right) = -\frac{1}{y^2} f_2' - \frac{x}{y^2} f_{12}'' - \frac{x}{y^3} f_{22}''$。

$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(-\frac{x}{y^2} f_2'\right) = \frac{2x}{y^3} f_2' - \frac{x}{y^2}\left[f_{21}'' \cdot 0 + f_{22}'' \cdot \left(-\frac{x}{y^2}\right)\right] = \frac{2x}{y^3} f_2' + \frac{x^2}{y^4} f_{22}''$。

(3) $z = f(\cos x, \cos y)$。

解：$\frac{\partial z}{\partial x} = f_1' \cdot (-\sin x) + f_2' \cdot 0 = -\sin x \cdot f_1'$。 $\frac{\partial z}{\partial y} = f_1' \cdot 0 + f_2' \cdot (-\sin y) = -\sin y \cdot f_2'$。

$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(-\sin x \cdot f_1'\right) = -\cos x \cdot f_1' - \sin x \cdot \left[f_{11}'' (-\sin x)\right] = -\cos x \cdot f_1' + \sin^2 x \cdot f_{11}''$。

$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial y}\left(-\sin x \cdot f_1'\right) = -\sin x \left(f_{11}'' \cdot 0 + f_{12}'' \cdot (-\sin y)\right) = \sin x \sin y \cdot f_{12}''$。

$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(-\sin y \cdot f_2'\right) = -\cos y \cdot f_2' - \sin y\left[f_{21}'' \cdot 0 + f_{22}'' \cdot (-\sin y)\right] = -\cos y \cdot f_2' + \sin^2 y \cdot f_{22}''$。

(4) $z = f(x^2 y, x y^2)$

解：$\frac{\partial z}{\partial x} = f_1' \cdot 2xy + f_2' \cdot y^2 = 2xy \cdot f_1' + y^2 \cdot f_2'$。 $\frac{\partial z}{\partial y} = f_1' \cdot x^2 + 2xy \cdot f_2' = x^2 f_1' + 2xy \cdot f_2'$。

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(2xy\cdot f_1' + y^2 f_2'\right) = 2y\cdot f_1' + f_{11}''\cdot(2xy) + f_{12}''\cdot y^2 + y^2\cdot f_{21}''\cdot 2xy + y^2\cdot f_{22}''\cdot y^2$$

$$= 2y\cdot f_1' + 4x^2y^2 f_{11}'' + 4xy^3 f_{12}'' + y^4 f_{22}''$$

$$\frac{\partial^2 z}{\partial x\partial y} = \frac{\partial}{\partial x}\left(x^2 f_1' + 2xy\cdot f_2'\right) = 2x f_1' + x^2 f_{11}''\cdot 2xy + x^2 f_{12}''\cdot y^2 + 2y\cdot f_2' + 2xy\cdot f_{21}''\cdot 2xy + 2xy\cdot f_{22}''\cdot y^2$$

$$= 2x f_1' + 2y f_2' + 2x^3 y f_{11}'' + 5x^2y^2 f_{12}'' + 2xy^3 f_{22}''$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(x^2 f_1' + 2xy\cdot f_2'\right) = x^2 f_{11}''\cdot x^2 + x^2 f_{12}''\cdot 2xy + 2x\cdot f_2' + 2xy\cdot f_{21}''\cdot x^2 + 2xy\cdot f_{22}''\cdot 2xy$$

$$= 2x\cdot f_2' + 4x^2y^2 f_{22}'' + 4x^3 y f_{12}'' + x^4 f_{11}''$$

12. 设 f 具有二阶连续偏导数，$u = f(x+y, x-y, xy)$，求 $\frac{\partial^2 u}{\partial x^2}$，$\frac{\partial^2 u}{\partial x\partial y}$.

解：$\frac{\partial u}{\partial x} = f_1'\cdot 1 + f_2'\cdot 1 + f_3'\cdot y = f_1' + f_2' + y f_3'$.

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(f_1' + f_2' + y f_3'\right) = f_{11}''\cdot 1 + f_{12}''\cdot 1 + f_{13}''\cdot y + f_{21}''\cdot 1 + f_{22}''\cdot 1 + f_{23}''\cdot y + y f_{31}''\cdot 1 + y f_{32}''\cdot 1 + y f_{33}''\cdot y$$

$$= f_{11}'' + f_{22}'' + y^2 f_{33}'' + 2f_{12}'' + 2y f_{13}'' + 2y f_{23}''$$

$$\frac{\partial^2 u}{\partial x\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial y}\left(f_1' + f_2' + y f_3'\right)$$

$$= f_{11}''\cdot 1 + f_{12}''\cdot(-1) + f_{13}''\cdot x + f_{21}''\cdot 1 + f_{22}''\cdot(-1) + f_{23}''\cdot x + f_3' + y f_{31}''\cdot 1 + y f_{32}''\cdot(-1) + y f_{33}''\cdot x$$

$$= f_3' + f_{11}'' - f_{22}'' + xy f_{33}'' + (x+y) f_{13}'' + (x-y) f_{23}''$$

13. 设 $f(x,y) = \int_0^{xy} e^{-t^2}dt$，求 $\frac{x}{y}\frac{\partial^2 f}{\partial x^2} - 2\frac{\partial^2 f}{\partial x\partial y} + \frac{y}{x}\frac{\partial^2 f}{\partial y^2}$.

解：$\frac{\partial f}{\partial x} = y e^{-x^2y^2}$，$\frac{\partial f}{\partial y} = x e^{-x^2y^2}$.

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(y e^{-x^2y^2}\right) = y e^{-x^2y^2}\cdot(-2xy^2) = -2xy^3 e^{-x^2y^2}$$

$$\frac{\partial^2 f}{\partial x\partial y} = \frac{\partial}{\partial y}\left(y e^{-x^2y^2}\right) = e^{-x^2y^2} + y e^{-x^2y^2}\cdot(-2x^2y) = (1-2x^2y^2)e^{-x^2y^2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(x e^{-x^2y^2}\right) = x e^{-x^2y^2}\cdot(-2x^2y) = -2x^3y e^{-x^2y^2}$$

$$\frac{x}{y}\frac{\partial^2 f}{\partial x^2} - 2\frac{\partial^2 f}{\partial x\partial y} + \frac{y}{x}\frac{\partial^2 f}{\partial y^2} = \frac{x}{y}(-2xy^3)e^{-x^2y^2} - 2(1-2x^2y^2)e^{-x^2y^2} + \frac{y}{x}(-2x^3y)e^{-x^2y^2}$$

$$= (-2x^2y^2 - 2 + 4x^2y^2 - 2x^2y^2)e^{-x^2y^2} = -2e^{-x^2y^2}$$

14. 设 $u=f(x,y)$ 具有连续二阶连续偏导数，$x=\frac{1}{2}(s-\sqrt{3}t)$，$y=\frac{1}{2}(\sqrt{3}s+t)$.

证明：(1) $\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=\left(\frac{\partial u}{\partial s}\right)^2+\left(\frac{\partial u}{\partial t}\right)^2$.

(2) $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=\frac{\partial^2 u}{\partial s^2}+\frac{\partial^2 u}{\partial t^2}$.

证明：(1) $\frac{\partial u}{\partial s}=\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial s}+\frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial s}=\frac{\partial u}{\partial x}\cdot\frac{1}{2}+\frac{\partial u}{\partial y}\cdot\frac{\sqrt{3}}{2}$

$\frac{\partial u}{\partial t}=\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial t}+\frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial t}=\frac{\partial u}{\partial x}\cdot\left(-\frac{\sqrt{3}}{2}\right)+\frac{\partial u}{\partial y}\cdot\frac{1}{2}$.

$\therefore\ \left(\frac{\partial u}{\partial s}\right)^2+\left(\frac{\partial u}{\partial t}\right)^2=\left(\frac{1}{2}\frac{\partial u}{\partial x}+\frac{\sqrt{3}}{2}\frac{\partial u}{\partial y}\right)^2+\left(-\frac{\sqrt{3}}{2}\frac{\partial u}{\partial x}+\frac{1}{2}\frac{\partial u}{\partial y}\right)^2=\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2$.

(2) $\frac{\partial^2 u}{\partial s^2}=\frac{\partial}{\partial s}\left(\frac{1}{2}\frac{\partial u}{\partial x}+\frac{\sqrt{3}}{2}\frac{\partial u}{\partial y}\right)=\frac{1}{2}\left(\frac{\partial^2 u}{\partial x^2}\cdot\frac{1}{2}+\frac{\partial^2 u}{\partial x\partial y}\cdot\frac{\sqrt{3}}{2}\right)+\frac{\sqrt{3}}{2}\left(\frac{\partial^2 u}{\partial x\partial y}\cdot\frac{1}{2}+\frac{\partial^2 u}{\partial y^2}\cdot\frac{\sqrt{3}}{2}\right)$

$=\frac{1}{4}\frac{\partial^2 u}{\partial x^2}+\frac{\sqrt{3}}{2}\frac{\partial^2 u}{\partial x\partial y}+\frac{3}{4}\frac{\partial^2 u}{\partial y^2}$.

$\frac{\partial^2 u}{\partial t^2}=\frac{\partial}{\partial t}\left(-\frac{\sqrt{3}}{2}\frac{\partial u}{\partial x}+\frac{1}{2}\frac{\partial u}{\partial y}\right)=-\frac{\sqrt{3}}{2}\left(\frac{\partial^2 u}{\partial x^2}\cdot\frac{-\sqrt{3}}{2}+\frac{\partial^2 u}{\partial x\partial y}\cdot\frac{1}{2}\right)+\frac{1}{2}\left(\frac{\partial^2 u}{\partial x\partial y}\cdot\frac{-\sqrt{3}}{2}+\frac{\partial^2 u}{\partial y^2}\cdot\frac{1}{2}\right)$

$=+\frac{3}{4}\frac{\partial^2 u}{\partial x^2}-\frac{\sqrt{3}}{2}\frac{\partial^2 u}{\partial x\partial y}+\frac{1}{4}\frac{\partial^2 u}{\partial y^2}$.

$\therefore\ \frac{\partial^2 u}{\partial s^2}+\frac{\partial^2 u}{\partial t^2}=\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}$.

15. 设有映射：$f:\ R^2\to R^3,\quad (x,y)^T\mapsto(u,v,w)^T$.

$g:\ R^2\to R^2,\quad (s,t)^T\mapsto(x,y)^T$.

其中 $u=x+y,\quad v=xy,\quad w=\frac{y}{x},\quad x=\frac{s}{s^2+t^2},\quad y=\frac{t}{s^2+t^2}$.

求复合映射 $f\circ g$ 的 Jacobi 矩阵.

解：
$$J(f\circ g)=J(f)\cdot J(g)=\begin{bmatrix}\frac{\partial u}{\partial x}&\frac{\partial u}{\partial y}\\ \frac{\partial v}{\partial x}&\frac{\partial v}{\partial y}\\ \frac{\partial w}{\partial x}&\frac{\partial w}{\partial y}\end{bmatrix}\cdot\begin{bmatrix}\frac{\partial x}{\partial s}&\frac{\partial x}{\partial t}\\ \frac{\partial y}{\partial s}&\frac{\partial y}{\partial t}\end{bmatrix}$$

$$=\begin{bmatrix}1&1\\ y&x\\ -\frac{y}{x^2}&\frac{1}{x}\end{bmatrix}\begin{bmatrix}\frac{s^2+t^2-2s^2}{(s^2+t^2)^2}&\frac{-2st}{(s^2+t^2)^2}\\ \frac{-2st}{(s^2+t^2)^2}&\frac{s^2+t^2-2t^2}{(s^2+t^2)^2}\end{bmatrix}=\frac{1}{(s^2+t^2)^2}\begin{bmatrix}1&1\\ y&x\\ -\frac{y}{x^2}&\frac{1}{x}\end{bmatrix}\begin{bmatrix}t^2-s^2&-2st\\ -2st&s^2-t^2\end{bmatrix}$$

$$= \frac{1}{(s^2+t^2)^2}\begin{bmatrix} -s^2-2st+t^2 & s^2-2st-t^2 \\ -y(s^2-t^2)-2xst & x(s^2-t^2)-2yst \\ \frac{y}{x^2}(s^2-t^2)-\frac{2}{x}st & \frac{1}{x}(s^2-t^2)+\frac{y}{x^2}\cdot 2st \end{bmatrix}$$

16. 设映射 $\vec{f}=(f_1,f_2)$, $\vec{g}=(g_1,g_2)$, 其中

$$f_1(s,t)=s^2+t^2, \quad f_2(s,t)=s^2-t^2.$$

$$g_1(x,y)=\ln\sqrt{x^2+y^2}, \quad g_2(x,y)=\arctan\frac{y}{x}.$$

求复合映射 $\vec{f}\circ\vec{g}$ 的 Jacobi 矩阵.

解:

$$J(\vec{f}\circ\vec{g}) = J(\vec{f})\cdot J(\vec{g}) = \begin{bmatrix} \frac{\partial f_1}{\partial s} & \frac{\partial f_1}{\partial t} \\ \frac{\partial f_2}{\partial s} & \frac{\partial f_2}{\partial t} \end{bmatrix}\cdot\begin{bmatrix} \frac{\partial g_1}{\partial x} & \frac{\partial g_1}{\partial y} \\ \frac{\partial g_2}{\partial x} & \frac{\partial g_2}{\partial y} \end{bmatrix}$$

$$= \begin{bmatrix} 2s & 2t \\ 2s & -2t \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{x^2+y^2}}\cdot\frac{x}{\sqrt{x^2+y^2}} & \frac{1}{\sqrt{x^2+y^2}}\cdot\frac{y}{\sqrt{x^2+y^2}} \\ \frac{1}{1+(\frac{y}{x})^2}\cdot(-\frac{y}{x^2}) & \frac{1}{1+(\frac{y}{x})^2}\cdot\frac{1}{x} \end{bmatrix}$$

$$= 2\begin{bmatrix} s & t \\ s & -t \end{bmatrix}\begin{bmatrix} \frac{x}{x^2+y^2} & \frac{y}{x^2+y^2} \\ -\frac{y}{x^2+y^2} & \frac{x}{x^2+y^2} \end{bmatrix} = \frac{2}{x^2+y^2}\begin{bmatrix} s & t \\ s & -t \end{bmatrix}\begin{bmatrix} x & y \\ -y & x \end{bmatrix}$$

$$= \frac{2}{x^2+y^2}\begin{bmatrix} sx-ty & sy+tx \\ sx+ty & sy-tx \end{bmatrix}.$$

17. 设在直角坐标系 (x,y) 下, 实函数 u, v 满足 Cauchy-Riemann 方程:

$$u_x' = v_y', \quad u_y' = -v_x'.$$

证明在极坐标系 (r,θ) 下, 上述方程相应地变成 $u_r' = \frac{1}{r}v_\theta'$, $\frac{1}{r}u_\theta' = -v_r'$.

证明: $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$

$$u_r' = u_x'\cdot\frac{\partial x}{\partial r} + u_y'\cdot\frac{\partial y}{\partial r} = \cos\theta\cdot u_x' + \sin\theta\cdot u_y' \quad ①$$

$$u_\theta' = u_x'\cdot\frac{\partial x}{\partial\theta} + u_y'\cdot\frac{\partial y}{\partial\theta} = -r\sin\theta\cdot u_x' + r\cos\theta\cdot u_y' \quad ②$$

$$v_r' = u_x'\cdot\frac{\partial x}{\partial r} + u_y'\cdot\frac{\partial y}{\partial r} = \cos\theta\cdot v_x' + \sin\theta\cdot v_y' \quad ③$$

$$v_\theta' = v_x'\cdot\frac{\partial x}{\partial\theta} + v_y'\cdot\frac{\partial y}{\partial\theta} = -r\sin\theta\cdot v_x' + r\cos\theta\cdot v_y' \quad ④.$$

由①.②解得: $\begin{cases} u'_x = \cos\theta \cdot u'_r - \frac{\sin\theta}{r} u'_\theta \\ u'_y = \sin\theta \cdot u'_r + \frac{\cos\theta}{r} u'_\theta \end{cases}$

由③.④解得: $\begin{cases} u'_x = \sin\theta \cdot v'_r + \frac{\cos\theta}{r} v'_\theta \\ u'_y = -\cos\theta \cdot v'_r + \frac{\sin\theta}{r} v'_\theta \end{cases}$

$\therefore \begin{cases} \cos\theta \cdot u'_r - \frac{\sin\theta}{r} u'_\theta = \sin\theta \cdot v'_r + \frac{\cos\theta}{r} v'_\theta \\ \sin\theta \cdot u'_r + \frac{\cos\theta}{r} u'_\theta = -\cos\theta \cdot v'_r + \frac{\sin\theta}{r} v'_\theta \end{cases}$ 即: $\begin{cases} u'_r = \frac{1}{r} v'_\theta \\ \frac{1}{r} u'_\theta = -v'_r \end{cases}$

18. 求 λ 和 μ, 使得线性变换 $\begin{cases} \xi = x + \lambda y \\ \eta = x + \mu y \end{cases}$

将微分关系式 $A\frac{\partial^2 u}{\partial x^2} + 2B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} = 0$ 化简为 $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$.

其中 A.B.C 为常数, 且 $C \neq 0$、$B^2 - AC > 0$.

解: $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta}\cdot\frac{\partial \eta}{\partial x} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}$.

$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi}\cdot\frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta}\cdot\frac{\partial \eta}{\partial y} = \lambda\frac{\partial u}{\partial \xi} + \mu\frac{\partial u}{\partial \eta}$.

$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}\right) = \left(\frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \xi \partial \eta}\right)\cdot 1 + \left(\frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}\right)\cdot 1 = \frac{\partial^2 u}{\partial \xi^2} + 2\frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}$.

$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}\right) = \left(\frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \xi \partial \eta}\right)\cdot\lambda + \left(\frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}\right)\cdot\mu = \lambda\frac{\partial^2 u}{\partial \xi^2} + (\lambda+\mu)\frac{\partial^2 u}{\partial \xi \partial \eta} + \mu\frac{\partial^2 u}{\partial \eta^2}$.

$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\lambda\frac{\partial u}{\partial \xi} + \mu\frac{\partial u}{\partial \eta}\right) = \lambda\left(\lambda\frac{\partial^2 u}{\partial \xi^2} + \mu\frac{\partial^2 u}{\partial \xi \partial \eta}\right) + \mu\left(\lambda\frac{\partial^2 u}{\partial \xi \partial \eta} + \mu\frac{\partial^2 u}{\partial \eta^2}\right) = \lambda^2\frac{\partial^2 u}{\partial \xi^2} + 2\lambda\mu\frac{\partial^2 u}{\partial \xi \partial \eta} + \mu^2\frac{\partial^2 u}{\partial \eta^2}$.

$A\frac{\partial^2 u}{\partial x^2} + 2B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} = \left(A + 2\lambda B + C\lambda^2\right)\frac{\partial^2 u}{\partial \xi^2} + \left[2A + 2(\lambda+\mu)B + 2\lambda\mu C\right]\frac{\partial^2 u}{\partial \xi \partial \eta}$

$+ \left(A + 2\mu B + C\mu^2\right)\frac{\partial^2 u}{\partial \eta^2} = 0$.

当 $\begin{cases} A + 2\lambda B + \lambda^2 = 0 \\ A + 2\mu B + \mu^2 = 0 \end{cases}$ 时, 上式化简为 $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$.

$\therefore$ λ, μ 为 $A + 2tB + Ct^2 = 0$ 的二个根

当 $C \neq 0$, $B^2 - AC > 0$ 时, 有:

$\lambda, \mu = -\frac{1}{C}\left(\sqrt{B^2 - AC} \pm B\right)$.

19. 设 $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$. 且 $\begin{cases} u = x+y \\ v = \frac{y}{x} \end{cases}$ $w = \frac{z}{x}$.

写为新的因变量 w 关于新的自变量 u.v 所满足的微分关系式

解: $z = wx$.

$$\frac{\partial z}{\partial x} = w + x\cdot\frac{\partial w}{\partial x} = w + x\cdot\left(\frac{\partial w}{\partial u} - \frac{y}{x^2}\frac{\partial w}{\partial v}\right) = w + x\frac{\partial w}{\partial u} - \frac{y}{x}\frac{\partial w}{\partial v}.$$

$$\frac{\partial z}{\partial y} = x\frac{\partial w}{\partial y} = x\left(\frac{\partial w}{\partial u} + \frac{1}{x}\frac{\partial w}{\partial v}\right) = x\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v}.$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(w + x\frac{\partial w}{\partial u} - \frac{y}{x}\frac{\partial w}{\partial v}\right) = \frac{\partial w}{\partial x} + \frac{\partial w}{\partial u} + x\frac{\partial}{\partial x}\left(\frac{\partial w}{\partial u}\right) + \frac{y}{x^2}\frac{\partial w}{\partial v} - \frac{y}{x}\frac{\partial}{\partial x}\left(\frac{\partial w}{\partial v}\right)$$

$$= 2\frac{\partial w}{\partial u} - \frac{y}{x^2}\frac{\partial w}{\partial v} + x\left(\frac{\partial^2 w}{\partial u^2} - \frac{y}{x^2}\frac{\partial^2 w}{\partial u \partial v}\right) + \frac{y}{x^2}\frac{\partial w}{\partial v} - \frac{y}{x}\left(\frac{\partial^2 w}{\partial u \partial v} - \frac{y}{x^2}\frac{\partial^2 w}{\partial v^2}\right)$$

$$= 2\frac{\partial w}{\partial u} + x\frac{\partial^2 w}{\partial u^2} - \frac{2y}{x}\frac{\partial^2 w}{\partial u \partial v} + \frac{y^2}{x^3}\frac{\partial^2 w}{\partial v^2}.$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(x\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v}\right) = \frac{\partial w}{\partial u} + x\frac{\partial}{\partial x}\left(\frac{\partial w}{\partial u}\right) + \frac{\partial}{\partial x}\left(\frac{\partial w}{\partial v}\right)$$

$$= \frac{\partial w}{\partial u} + x\left(\frac{\partial^2 w}{\partial u^2} - \frac{y}{x^2}\frac{\partial^2 w}{\partial u \partial v}\right) + \frac{\partial^2 w}{\partial u \partial v} - \frac{y}{x^2}\frac{\partial^2 w}{\partial v^2}$$

$$= \frac{\partial w}{\partial u} + x\frac{\partial^2 w}{\partial u^2} + \left(1 - \frac{y}{x}\right)\frac{\partial^2 w}{\partial u \partial v} - \frac{y}{x^2}\frac{\partial^2 w}{\partial v^2}.$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(x\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v}\right) = x\frac{\partial^2 w}{\partial u^2} + \frac{\partial^2 w}{\partial u \partial v} + \frac{1}{x}\left(x\frac{\partial^2 w}{\partial u \partial v} + \frac{\partial^2 w}{\partial v^2}\right)$$

$$= x\frac{\partial^2 w}{\partial u^2} + 2\frac{\partial^2 w}{\partial u \partial v} + \frac{1}{x}\frac{\partial^2 w}{\partial v^2}.$$

则有: $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \left(\frac{y^2}{x^3} + \frac{1}{x} + \frac{2y}{x^2}\right)\frac{\partial^2 w}{\partial v^2} = \frac{(x+y)^2}{x^3}\cdot\frac{\partial^2 w}{\partial v^2} = 0$.

$\therefore$ w 关于 u.v 满足: $\frac{\partial^2 w}{\partial v^2} = 0$.

§4. 隐函数微分法及其应用

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1. 求下列隐函数的导数 $\frac{dy}{dx}$.

(2) $e^x + \sin(x+y) + xy = 0$.

解: $e^x + \cos(x+y)\cdot\left(\frac{dy}{dx} + 1\right) + x\cdot\frac{dy}{dx} + y = 0 \Rightarrow \frac{dy}{dx} = -\frac{y + e^x + \cos(x+y)}{x + \cos(x+y)}$.

(2) $e^x\cos y + e^y \sin x = 1$.

解: $e^x\cos y + e^x\cdot(-\sin y)\cdot\frac{dy}{dx} + e^y\cdot\frac{dy}{dx}\cdot\sin x + e^y\cos x = 0$.

$$\Rightarrow \frac{dy}{dx} = \frac{e^x\cos y + e^y\cos x}{e^x\sin y - e^y\sin x}.$$

2. 求下列隐函数的偏导数 z'_x, z'_y.

(1) $yz^2 - xz + xy - 4z^3 = 0$.

解: $z'_x = -\frac{F'_x}{F'_z} = -\frac{-z+y}{2yz - x - 12z^2} = \frac{z-y}{2yz-x-12z^2}$.

$$z'_y = -\frac{F'_y}{F'_z} = -\frac{z^2+x}{2yz-x-12z^2} = -\frac{z^2+x}{2yz-x-12z^2}.$$

(2) $\cos^2 x + \cos^2 y + \cos^2 z = 1$.

解: $z'_x = -\frac{F'_x}{F'_z} = -\frac{2\cos x\cdot(-\sin x)}{2\cos z\cdot(-\sin z)} = -\frac{\cos x\cdot\sin x}{\cos z\cdot\sin z}$.

$$z'_y = -\frac{F'_y}{F'_z} = -\frac{2\cos y\cdot(-\sin y)}{2\cos z\cdot(-\sin z)} = -\frac{\cos y\cdot\sin y}{\cos z\cdot\sin z}.$$

(3) $x + 2y + z - 2\sqrt{xyz} = 0$.

解: $z'_x = -\frac{F'_x}{F'_z} = -\frac{1-2\cdot\frac{yz}{2\sqrt{xyz}}}{1-2\cdot\frac{xy}{2\sqrt{xyz}}} = -\frac{\sqrt{xyz}-yz}{\sqrt{xyz}-xy}$.

$$z'_y = -\frac{F'_y}{F'_z} = -\frac{2-2\cdot\frac{xz}{2\sqrt{xyz}}}{1-2\cdot\frac{xy}{2\sqrt{xyz}}} = -\frac{2\sqrt{xyz}-xz}{\sqrt{xyz}-xy}.$$

(4) $\frac{x}{z} = \ln\frac{z}{y}$. $\left(\Rightarrow \frac{x}{z} - \ln\frac{z}{y} = F(x,y,z) = 0\right)$.

解: $z'_x = -\frac{F'_x}{F'_z} = -\frac{\frac{1}{z}}{-\frac{x}{z^2}-\frac{y}{z}\cdot\frac{1}{y}} = \frac{z}{x+z}$.

$$z'_y = -\frac{F'_y}{F'_z} = -\frac{-\frac{y}{z}\cdot\left(-\frac{z}{y^2}\right)}{-\frac{x}{z^2}-\frac{y}{z}\cdot\frac{1}{y}} = \frac{z^2}{y(x+z)}.$$

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3. 设 F 是三元可微函数 $F(x, x+y, x+y+z)=0$ 求 z'_x, z'_y.

解: $z'_x=-\dfrac{F'_x}{F'_z}=-\dfrac{F'_1+F'_2+F'_3}{F'_3}$ $\quad z'_y=-\dfrac{F'_y}{F'_z}=-\dfrac{F'_2+F'_3}{F'_3}$

4. 设 $x=x(y,z)$, $y=y(x,z)$, $z=z(x,y)$ 都是由 $F(x,y,z)=0$ 所确定的具有连续偏导数的函数. 证明: $\dfrac{\partial x}{\partial y}\cdot\dfrac{\partial y}{\partial z}\cdot\dfrac{\partial z}{\partial x}=-1$.

证明: $\dfrac{\partial x}{\partial y}=-\dfrac{F'_y}{F'_x}$ $\quad \dfrac{\partial y}{\partial z}=-\dfrac{F'_z}{F'_y}$ $\quad \dfrac{\partial z}{\partial x}=-\dfrac{F'_x}{F'_z}$.

$\therefore \dfrac{\partial x}{\partial y}\cdot\dfrac{\partial y}{\partial z}\cdot\dfrac{\partial z}{\partial x}=\left(-\dfrac{F'_y}{F'_x}\right)\left(-\dfrac{F'_z}{F'_y}\right)\left(-\dfrac{F'_x}{F'_z}\right)=-1$.

5. 设 $z^3-3xyz=a^3$, 求 $\dfrac{\partial^2 z}{\partial x\partial y}$.

解: $F(x,y,z)=z^3-3xyz=a^3$.

$$\frac{\partial z}{\partial x}=-\frac{F'_x}{F'_z}=-\frac{-3yz}{3z^2-3xy}=\frac{yz}{z^2-xy}.$$

$$\frac{\partial z}{\partial y}=-\frac{F'_y}{F'_z}=-\frac{-3xz}{3z^2-3xy}=\frac{xz}{z^2-xy}.$$

$$\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial y}\left(\frac{yz}{z^2-xy}\right)=\frac{1}{(z^2-xy)^2}\left[(z^2-xy)\cdot\left(z+y\frac{\partial z}{\partial y}\right)-yz\left(2z\frac{\partial z}{\partial y}-x\right)\right]$$

$$=\frac{1}{(z^2-xy)^2}\left[(z^2-xy)\left(z+\frac{xyz}{z^2-xy}\right)-2z^2\cdot\frac{xyz}{z^2-xy}+xyz\right]$$

$$=\frac{1}{(z^2-xy)^2}\left[z^3-xyz+z^2\cdot\frac{xyz}{z^2-xy}-xy\cdot\frac{xyz}{z^2-xy}-2z^2\cdot\frac{xyz}{z^2-xy}+xyz\right]$$

$$=\frac{1}{(z^2-xy)^2}\left[z^3+xyz-2z^2\cdot\frac{xyz}{z^2-xy}\right]=\frac{z^5-2xyz^3-x^2y^2z}{(z^2-xy)^3}$$

6. 设 $e^z-xyz=0$, 求 $\dfrac{\partial^2 z}{\partial x^2}$.

解: $\dfrac{\partial z}{\partial x}=-\dfrac{F'_x}{F'_z}=-\dfrac{-yz}{e^z-xy}=\dfrac{yz}{e^z-xy}$.

$$\frac{\partial^2 z}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{yz}{e^z-xy}\right)=\frac{1}{(e^z-xy)^2}\left[(e^z-xy)\cdot y\frac{\partial z}{\partial x}-yz\left(e^z\cdot\frac{\partial z}{\partial x}-y\right)\right]$$

$$= \frac{y}{e^z - xy}\left(\frac{yz}{e^z-xy}\right) - \frac{yz\cdot e^z}{(e^z-xy)^2}\left(\frac{yz}{e^z-xy}\right) + \frac{y^2z}{(e^z-xy)^2}$$

$$= \frac{2y^2z}{(e^z-xy)^2} - \frac{y^2z^2e^z}{(e^z-xy)^3} = \frac{y^2z(2e^z-2xy-ze^z)}{(e^z-xy)^3}.$$

7. 试求由下列方程组确定的映射 $f: x \mapsto \begin{pmatrix} y \\ z \end{pmatrix}$ 的 Jacobi 矩阵.

(1) $\begin{cases} z = x^2+y^2 \\ x^2+2y^2+3z^2 = 20. \end{cases}$

解: $\begin{cases} F_1(x,y,z) = x^2+y^2-z = 0. \\ F_2(x,y,z) = x^2+2y^2+3z^2-20 = 0. \end{cases}$

$$\begin{bmatrix} \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z} \\ \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z} \end{bmatrix}^{-1} = \begin{bmatrix} 2y & -1 \\ 4y & 6z \end{bmatrix} = \frac{1}{12yz+4y}\begin{bmatrix} +6z & 1 \\ -4y & +2y \end{bmatrix} = \frac{1}{4y(3z+1)}\begin{bmatrix} 6z & 1 \\ -4y & 2y \end{bmatrix}$$

$$J = \begin{pmatrix} \frac{\partial y}{\partial x} \\ \frac{\partial z}{\partial x} \end{pmatrix} = -\begin{bmatrix} \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z} \\ \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z} \end{bmatrix}^{-1}\begin{bmatrix} \frac{\partial F_1}{\partial x} \\ \frac{\partial F_2}{\partial x} \end{bmatrix} = \frac{-1}{4y(3z+1)}\begin{bmatrix} 6z & 1 \\ -4y & 2y \end{bmatrix}\begin{pmatrix} 2x \\ -2x \end{pmatrix}$$

$$= \frac{-1}{4y(3z+1)}\begin{pmatrix} 2x(6z+1) \\ 2x(-4y+2y) \end{pmatrix} = \frac{-x}{2y(3z+1)}\begin{pmatrix} 6z+1 \\ -2y \end{pmatrix} = \begin{pmatrix} -\frac{x(6z+1)}{y(6z+2)} \\ \frac{x}{3z+1} \end{pmatrix}.$$

(2) $\begin{cases} x+y+z = 0 \\ x^2+y^2+z^2 = 1 \end{cases}$

解: $\begin{cases} F_1(x,y,z) = x+y+z = 0. \\ F_2(x,y,z) = x^2+y^2+z^2-1 = 0. \end{cases}$ $\begin{bmatrix} \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z} \\ \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z} \end{bmatrix}^{-1} = \begin{pmatrix} 1 & 1 \\ 2y & 2z \end{pmatrix}^{-1} = \frac{1}{2(z-y)}\begin{pmatrix} 2z & -1 \\ -2y & 1 \end{pmatrix}$

$$J = \begin{pmatrix} \frac{\partial y}{\partial x} \\ \frac{\partial z}{\partial x} \end{pmatrix} = -\begin{bmatrix} \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z} \\ \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z} \end{bmatrix}^{-1}\begin{pmatrix} \frac{\partial F_1}{\partial x} \\ \frac{\partial F_2}{\partial x} \end{pmatrix} = \frac{-1}{2(z-y)}\begin{pmatrix} 2z & -1 \\ -2y & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 2x \end{pmatrix}$$

$$= \frac{-1}{2(z-y)}\begin{pmatrix} 2z-2x \\ -2y+2x \end{pmatrix} = \begin{pmatrix} -\frac{z-x}{z-y} \\ -\frac{x-y}{z-y} \end{pmatrix} = \begin{pmatrix} \frac{x-z}{z-y} \\ \frac{x-y}{y-z} \end{pmatrix}.$$

8. 求由下列方程组确定的映射 $(x,y)^T \mapsto (u,v)^T$ 的 Jacobi 阵.

(1) $\begin{cases} x = e^u + u\sin v \\ y = e^u - u\cos v \end{cases}$

解:
$$J(x,y) = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}^{-1} = \begin{bmatrix} e^u+\sin v & u\cos v \\ e^u-\cos v & u\sin v \end{bmatrix}^{-1}$$

$$= \frac{1}{u\sin v\,(e^u+\sin v) - u\cos v\,(e^u-\cos v)} \begin{bmatrix} u\sin v & -u\cos v \\ -(e^u-\cos v) & e^u+\sin v \end{bmatrix}$$

$$= \frac{1}{u + ue^u(\sin v - \cos v)} \begin{bmatrix} u\sin v & -u\cos v \\ \cos v - e^u & \sin v + e^u \end{bmatrix}$$

(2) $\begin{cases} u = f(ux, v+y) \\ v = g(u-x, v^2y) \end{cases}$

解: $\begin{cases} F_1(x,y,u,v) = f(ux, v+y) - u = 0. \\ F_2(x,y,u,v) = g(u-x, v^2y) - v = 0. \end{cases}$

$$J(x,y) = -\begin{bmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{bmatrix}^{-1} \begin{bmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{bmatrix} = -\begin{bmatrix} xf_1'-1 & f_2' \\ g_1' & 2vyg_2'-1 \end{bmatrix}^{-1} \begin{bmatrix} uf_1' & f_2' \\ -g_1' & v^2g_2' \end{bmatrix}$$

$$= \frac{-1}{(xf_1'-1)(2vyg_2'-1) - f_2'g_1'} \begin{bmatrix} 2vyg_2'-1 & -f_2' \\ -g_1' & xf_1'-1 \end{bmatrix} \begin{bmatrix} -uf_1' & f_2' \\ -g_1' & v^2g_2' \end{bmatrix}$$

$$= \frac{1}{g_1'f_2' - (xf_1'-1)(2vyg_2'-1)} \begin{bmatrix} 2uvyf_1'g_2' - uf_1' + f_2'g_1' & v(2y-v)f_2'g_2' - f_2' \\ -g_1'[1-(x+u)f_1'] & = f_2'g_1' + v^2xf_1'g_2' - v^2g_2' \end{bmatrix}$$

9. 设 $y = f(x,t)$, $F(x,y,t) = 0$, 其中二元函数 f 和三元函数 F 均具有连续一阶偏导数, 且 $F_t' \neq 0$. 证明: $\frac{dy}{dx} = \frac{f_x'F_t' - f_tF_x'}{f_tF_y' + F_t'}$.

证明: 由方程 $F(x,y,t) = 0$ 所确定的隐函数为 $t = h(x,y)$.

由 $y = f(x,t) = f(x, h(x,y))$ 确定了隐函数 $y = y(x)$.

同理，有：$t = h(x,y) = h(x, y(x)) = t(x)$.

将 $F(x,y,t)=0$ 两边对 x 求导，得：

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\cdot\frac{dy}{dx} + \frac{\partial F}{\partial t}\cdot\frac{dt}{dx} = 0 \quad \Rightarrow \quad \frac{dt}{dx} = -\frac{F'_x + F'_y\cdot\frac{dy}{dx}}{F'_t} \quad (F'_t \neq 0).$$

将 $y = f(x, t(x))$ 两边对 x 求导，得：

$$\frac{dy}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial t}\cdot\frac{dt}{dx} = f'_x + f'_t\cdot\left(-\frac{F'_x + F'_y\cdot\frac{dy}{dx}}{F'_t}\right)$$

$$= f'_x = \frac{f'_t}{F'_t}\left(F'_x + F'_y\cdot\frac{dy}{dx}\right).$$

$\therefore$ 有：$$\frac{dy}{dx} = \frac{f'_x F'_t - f'_t F'_x}{f'_t F'_y + F'_t}.$$

10. 求曲面 $e^z - z + xy = 3$ 在点 $(2,1,0)$ 处的切平面方程.

解：$F(x,y,z) = e^z - z + xy - 3$.

在 $(2,1,0)$ 处切平面法向 $\vec{n} = (F'_x, F'_y, F'_z)\big|_{P_0} = (y, x, e^z - 1)\big|_{P_0} = (1,2,0)$.

切平面方程：$1\times(x-2) + 2(y-1) + 0\cdot(z-0) = 0$.

即：$x + 2y - 4 = 0$.

11. 求椭球面 $x^2 + 2y^2 + z^2 = 1$ 上平行于平面 $x - y + 2z = 0$ 的切平面方程.

解：$\vec{n} = (F'_x, F'_y, F'_z) = (2x, 4y, 2z) = (1, -1, 2)t$ （平行于已知平面）.

即：$\frac{2x}{1} = \frac{4y}{-1} = \frac{2z}{2} = t$. $\Rightarrow$ $\begin{cases} x_0 = \frac{1}{2}t \\ y_0 = -\frac{1}{4}t \\ z_0 = t \end{cases}$

代入椭球面方程：$x_0^2 + 2y_0^2 + z_0^2 = 1$. 即有：

$$\frac{1}{4}t^2 + \frac{1}{8}t^2 + t^2 = 1. \quad \Rightarrow \quad t = \pm\sqrt{\frac{8}{11}}.$$

切平面方程：$(x - \frac{1}{2}t) - (y + \frac{1}{4}t) + 2(z - t) = 0$. $\Rightarrow$ $x - y + 2z - \frac{11}{4}t = 0$.

即：$x - y + 2z \pm \sqrt{\frac{11}{2}} = 0$.

12. 证明：曲面 $\sqrt{x} + \sqrt{y} + \sqrt{z} = \sqrt{a}$ $(a>0)$ 上任何点处的切平面在各坐标轴上的截距之和等于 a.

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证明：切平面 $\vec{n}=(F'_x\ F'_y\ F'_z)=\frac{1}{2}\left(\frac{1}{\sqrt{x}}\ \frac{1}{\sqrt{y}}\ \frac{1}{\sqrt{z}}\right)$，过点 (x_0,y_0,z_0) 切平面方程：

$$\frac{1}{\sqrt{x_0}}(x-x_0)+\frac{1}{\sqrt{y_0}}(y-y_0)+\frac{1}{\sqrt{z_0}}(z-z_0)=0.$$

$$\Rightarrow \frac{x}{\sqrt{x_0}}+\frac{y}{\sqrt{y_0}}+\frac{z}{\sqrt{z_0}}=\sqrt{x_0}+\sqrt{y_0}+\sqrt{z_0}=\sqrt{a}.$$

$$\Rightarrow \frac{x}{\sqrt{ax_0}}+\frac{y}{\sqrt{ay_0}}+\frac{z}{\sqrt{az_0}}=1.$$

即各坐标轴上截距之和为 $\sqrt{ax_0}+\sqrt{ay_0}+\sqrt{az_0}=\sqrt{a}\cdot\sqrt{a}=a$.

13. 设 $x=u+v$，$y=u^2+v^2$，$z=u^3+v^3$，求 z'_x，z'_y.

解：$(z'_x\ z'_y)=(z'_u\ z'_v)\begin{bmatrix}u'_x & u'_y\\ v'_x & v'_y\end{bmatrix}=(z'_u\ z'_v)\begin{bmatrix}x'_u & x'_v\\ y'_u & y'_v\end{bmatrix}^{-1}$

$$=(3u^2\ 3v^2)\begin{bmatrix}1 & 1\\ 2u & 2v\end{bmatrix}^{-1}=(3u^2\ 3v^2)\cdot\frac{1}{2v-2u}\begin{bmatrix}2v & -1\\ -2u & 1\end{bmatrix}$$

$$=\frac{3}{2(v-u)}(u^2\ v^2)\begin{bmatrix}2v & -1\\ -2u & 1\end{bmatrix}=\frac{3}{2(v-u)}\begin{bmatrix}2uv(u-v) & -u^2+v^2\end{bmatrix}$$

$$=\frac{3}{2(v-u)}\cdot(v-u)\begin{pmatrix}-2uv & u+v\end{pmatrix}=\frac{3}{2}\begin{pmatrix}-2uv & u+v\end{pmatrix}$$

由 $\begin{cases}x=u+v\\ y=u^2+v^2\end{cases}\Rightarrow uv=\frac{1}{2}(x^2-y)$. $\therefore\begin{cases}z'_x=-3uv=-\frac{3}{2}(x^2-y)\\ z'_y=\frac{3}{2}(u+v)=\frac{3}{2}x.\end{cases}$

14. 已知曲面 $x^2-y^2-3z=0$，求经过点 $A(0,0,-1)$ 且与直线 $\frac{x}{2}=\frac{y}{1}=\frac{z}{2}$ 平行的切平面方程.

解：切平面 $\vec{n}=(F'_x,F'_y,F'_z)=(2x,-2y,-3)$. $\vec{l}=(2,1,2)$.

设切点 $P(x_0,y_0,z_0)$. 则 $\vec{n}=(2x_0,-2y_0,-3)$.

切平面方程：$2x_0(x-x_0)-2y_0(y-y_0)-3(z-z_0)=0$.

即：$2x_0x-2y_0y-3z=2x_0^2-2y_0^2-3z_0$

$\because$ P点在曲面上，$\therefore$ 满足 $x_0^2-y_0^2-3z_0^2=0$.

$\therefore$ 切平面方程：$2x_0x-2y_0y-3z=3z_0$ 过 $A(0,0,-1)\Rightarrow z_0=1$.

由 $\vec{n}\cdot\vec{l}=0\Rightarrow\begin{cases}4x_0-2y_0-6=0\\ x_0^2-y_0^2=3z_0=3\end{cases}\Rightarrow\begin{cases}x_0=2\\ y_0=1\end{cases}$ $\therefore$ 切平面方程：

$4x-2y-3z-3=0$.

15. 设椭球面 $2x^2+3y^2+z^2=6$ 上点 $P(1,1,1)$ 处指向外侧的法向量为 $\vec{n}$，求函数 $u=\frac{\sqrt{6x^2+8y^2}}{z}$ 在点 P 处沿方向 $\vec{n}$ 的方向导数.

解: $\vec{n}=(F'_x,F'_y,F'_z)\big|_P=(4x,6y,2z)\big|_P=(4,6,2)$. $\vec{n}^0=\frac{1}{\sqrt{14}}(2,3,1)$.

$$u'_x\Big|_P=\frac{6x}{z\sqrt{6x^2+8y^2}}\Big|_P=\frac{6}{\sqrt{14}},\quad u'_y\Big|_P=\frac{8y}{z\sqrt{6x^2+8y^2}}\Big|_P=\frac{8}{\sqrt{14}},\quad u'_z\Big|_P=-\frac{\sqrt{6x^2+8y^2}}{z^2}\Big|_P=-\sqrt{14}.$$

$$\frac{\partial u}{\partial \vec{n}}=u'_x\cos\alpha_1+u'_y\cos\alpha_2+u'_z\cos\alpha_3$$

$$=\frac{6}{\sqrt{14}}\cdot\frac{2}{\sqrt{14}}+\frac{8}{\sqrt{14}}\cdot\frac{3}{\sqrt{14}}-\sqrt{14}\cdot\frac{1}{\sqrt{14}}=\frac{6}{7}+\frac{12}{7}-1=\frac{11}{7}.$$

16. 求曲线参数方程 $x=u\cos v$, $y=u\sin v$, $z=\sqrt{a^2-u^2}$. 分别求曲面在点 $A(x_0,y_0,z_0)$ 处的切平面方程，其中点 A 在曲面上.

解: 曲面方程: $z=\sqrt{a^2-x^2-y^2}$. 在 $A(x_0,y_0,z_0)$ 处的切平面法向 $\vec{n}$.

$$\vec{n}=(F'_x,F'_y,F'_z)=\left(\frac{x}{\sqrt{a^2-x^2-y^2}},\frac{y}{\sqrt{a^2-x^2-y^2}},1\right)_A=\left(\frac{x}{z},\frac{y}{z},1\right)_A$$

$$=(x_0,y_0,z_0)/z_0.\quad (\text{或曲面方程可改写为: } x^2+y^2+z^2=a^2).$$

∴ 切平面方程: $x_0(x-x_0)+y_0(y-y_0)+z_0(z-z_0)=0$.

即: $x_0x+y_0y+z_0z=x_0^2+y_0^2+z_0^2=a^2$.

$\Rightarrow x_0x+y_0y+z_0z=a^2$.

17. 试求空间曲线 $\begin{cases}x^2+y^2+z^2-3x=0\\2x-3y+5z-4=0\end{cases}$ 在 $(1,1,1)$ 处的切线方程.

解: $\begin{cases}F(x,y,z)=x^2+y^2+z^2-3x=0\\G(x,y,z)=2x-3y+5z-4=0\end{cases}$

$$\left|\begin{matrix}F'_y & F'_z\\ G'_y & G'_z\end{matrix}\right|_{P_0}=\left|\begin{matrix}2y & 2z\\ -3 & 5\end{matrix}\right|_{P_0}=16.\quad \left|\begin{matrix}F'_z & F'_x\\ G'_z & G'_x\end{matrix}\right|_{P_0}=\left|\begin{matrix}2z & 2x-3\\ 5 & 2\end{matrix}\right|_{P_0}=9.\quad \left|\begin{matrix}F'_x & F'_y\\ G'_x & G'_y\end{matrix}\right|_{P_0}=\left|\begin{matrix}2x-3 & 2y\\ 2 & -3\end{matrix}\right|_{P_0}=-1.$$

∴ $\vec{l}=(16,9,-1)$. 切线方程: $\frac{x-1}{16}=\frac{y-1}{9}=\frac{z-1}{-1}$.

18. 证明: 曲面 $f\left(\frac{x-a}{z-c},\frac{y-b}{z-c}\right)=0$ 上任一点处的切平面都过一定点.

解：设 $u=\frac{x-a}{z-c}$，$v=\frac{y-b}{z-c}$，则 $f(u,v)=0$。设 $P_0(x_0,y_0,z_0)$ 为曲面上任意一点，则该点处的法向量

$$\vec{n}=(-f'_x,-f'_y,f'_z)\Big|_{P_0}=\left(\frac{f'_u}{z_0-c},\frac{f'_v}{z_0-c},-\frac{(x_0-a)f'_u+(y_0-b)f'_v}{(z_0-c)^2}\right)$$

P_0 点处的切平面方程：

$$\frac{f'_u}{z_0-c}(x-x_0)+\frac{f'_v}{z_0-c}(y-y_0)-\frac{(x_0-a)f'_u+(y_0-b)f'_v}{(z_0-c)^2}(z-z_0)=0.$$

即：$$\left[(z_0-c)(x-x_0)-(x_0-a)(z-z_0)\right]f'_u+\left[(z_0-c)(y-y_0)-(y_0-b)(z-z_0)\right]f'_v=0.$$

且此点 (a,b,c) 满足上述的平面方程。

∴ 所有的平面均经过 (a,b,c) 点。

§5 方向导数、梯度

习题 P.52.

1. 求函数 $z=x^2+y^2$ 在点 $(1,2)$ 处沿从点 $(1,2)$ 到 $(2,2+\sqrt{3})$ 点方向的方向导数。

解：$\vec{l}=(2,2+\sqrt{3})-(1,2)=(1,\sqrt{3})$，$\vec{l}^0=\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$。

$$\frac{\partial z}{\partial \vec{l}}=\frac{\partial z}{\partial x}\cdot\cos\alpha_1+\frac{\partial z}{\partial y}\cdot\cos\alpha_2=2x\Big|_{(1,2)}\cdot\frac{1}{2}+2y\Big|_{(1,2)}\cdot\frac{\sqrt{3}}{2}=1+2\sqrt{3}.$$

2. 求函数 $u=xyz$ 在点 $(5,1,2)$ 处沿从该点到点 $(9,4,14)$ 方向的方向导数。

解：$\vec{l}=(9,4,14)-(5,1,2)=(4,3,12)$，$\vec{l}^0=\left(\frac{4}{13},\frac{3}{13},\frac{12}{13}\right)$。

$$\frac{\partial u}{\partial \vec{l}}=\frac{\partial u}{\partial x}\cdot\cos\alpha_1+\frac{\partial u}{\partial y}\cdot\cos\alpha_2+\frac{\partial u}{\partial z}\cos\alpha_3$$

$$=yz\Big|_{(5,1,2)}\cdot\frac{4}{13}+zx\Big|_{(5,1,2)}\cdot\frac{3}{13}+xy\Big|_{(5,1,2)}\cdot\frac{12}{13}=\frac{8}{13}+\frac{30}{13}+\frac{60}{13}=\frac{98}{13}.$$

3. 求函数 $u=\sum_{i,j=1}^{n}x_ix_j$ 在点 $(1,1,\cdots,1)$ 处沿 $\vec{l}=-(1,1,\cdots,1)$ 方向的方向导数。

解：$\vec{l}^0=-\frac{1}{\sqrt{n}}(1,1,\cdots,1)$，$\cos\alpha_i=-\frac{1}{\sqrt{n}}$，$i=1,2,\cdots,n$。

$$\frac{\partial u}{\partial \vec{l}}=\sum_{i=1}^{n}\frac{\partial u}{\partial x_i}\Big|_{(1,1,\cdots,1)}\cdot\cos\alpha_i=\sum_{j=1}^{n}\left[2\sum_{i=1}^{n}x_i\Big|_{(1,1,\cdots,1)}\cdot\left(-\frac{1}{\sqrt{n}}\right)\right]=\sum_{j=1}^{n}\left(-2\sqrt{n}\right)=-2n\sqrt{n}.$$

4. 已知 $u=x^2+y^2+z^2-xy+yz$，求 $P_0=(1,1,1)$，求 u 在点 P_0 处沿方向导数 $\frac{\partial u}{\partial \vec{l}}$ 的最大值和最小值，并指出相应的方向 $\vec{l}$.

解：$\frac{\partial u}{\partial \vec{l}} = \frac{\partial u}{\partial x}\cdot\cos\alpha_1 + \frac{\partial u}{\partial y}\cdot\cos\alpha_2 + \frac{\partial u}{\partial z}\cdot\cos\alpha_3$

$= (2x-y)\big|_{P_0}\cdot\cos\alpha_1 + (2y-x+z)\big|_{P_0}\cdot\cos\alpha_2 + (2z+y)\big|_{P_0}\cdot\cos\alpha_3$

$= \cos\alpha_1 + 2\cos\alpha_2 + 3\cos\alpha_3 = \operatorname{grad} u\cdot\vec{l}^0$. 即：$\operatorname{grad} u = (1,2,3)$.

$\max\left(\frac{\partial u}{\partial \vec{l}}\right) = \|\operatorname{grad} u\| = \sqrt{14}$. 此时 $\vec{l}_1^0 = \frac{\operatorname{grad} u}{\|\operatorname{grad} u\|} = \frac{1}{\sqrt{14}}(1,2,3)$.

$\min\left(\frac{\partial u}{\partial \vec{l}}\right) = -\|\operatorname{grad} u\| = -\sqrt{14}$. 此时 $\vec{l}_2^0 = -\frac{\operatorname{grad} u}{\|\operatorname{grad} u\|} = -\frac{1}{\sqrt{14}}(1,2,3)$.

当 $\operatorname{grad} u\cdot\vec{l}^0 = 0$ 即 $\operatorname{grad} u$ 与 $\vec{l}^0$ 垂直时，$\frac{\partial u}{\partial \vec{l}} = 0$.

设此时 $\vec{l}_3^0 = (r,s,t)$，则有：$r+2s+3t=0 \Rightarrow r=-2s-3t$.

$\therefore \vec{l}_3^0 = (-2s-3t)\vec{i} + s\vec{j} + t\vec{k}$.

5. 求下列数量场的梯度：

(1) $u=\sqrt{x^2+y^2}$.

解：$\operatorname{grad} u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right) = \left(\frac{x}{\sqrt{x^2+y^2}}, \frac{y}{\sqrt{x^2+y^2}}\right) = \frac{1}{\sqrt{x^2+y^2}}(x\vec{i}+y\vec{j})$.

(2) $u=\frac{xyz}{x+y+z}$.

解：$\operatorname{grad} u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right) = \frac{1}{(x+y+z)^2}\left(yz(y+z), xz(x+z), xy(x+y)\right)$

$= \frac{1}{(x+y+z)^2}\left(yz(y+z)\vec{i} + xz(x+z)\vec{j} + xy(x+y)\vec{k}\right)$.

(3) $u=\sum_{i=1}^{n} x_i$.

解：$\operatorname{grad} u = \left(\frac{\partial u}{\partial x_1}, \frac{\partial u}{\partial x_2}, \cdots, \frac{\partial u}{\partial x_n}\right) = (1,1,\cdots,1)$.

6. 设 $u=f(x,y,z)$ 具有连续的一阶偏导函数，沿方向 $\vec{l}(\cos\alpha,\cos\beta,\cos\gamma)$ 的二阶方向导数：$\frac{\partial^2 u}{\partial \vec{l}^2} = \frac{\partial}{\partial \vec{l}}\left(\frac{\partial u}{\partial \vec{l}}\right)$.

解: $\frac{\partial u}{\partial \vec{l}} = \frac{\partial f}{\partial x}\cos\alpha + \frac{\partial f}{\partial y}\cos\beta + \frac{\partial f}{\partial z}\cos\gamma$.

$$\frac{\partial^2 u}{\partial \vec{l}^2} = \frac{\partial}{\partial \vec{l}}\left(\frac{\partial u}{\partial \vec{l}}\right) = \frac{\partial}{\partial \vec{l}}\left(\frac{\partial f}{\partial x}\cos\alpha + \frac{\partial f}{\partial y}\cos\beta + \frac{\partial f}{\partial z}\cos\gamma\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\cos\alpha + \frac{\partial f}{\partial y}\cos\beta + \frac{\partial f}{\partial z}\cos\gamma\right)\cdot\cos\alpha$$

$$+ \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\cos\alpha + \frac{\partial f}{\partial y}\cos\beta + \frac{\partial f}{\partial z}\cos\gamma\right)\cdot\cos\beta$$

$$+ \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial x}\cos\alpha + \frac{\partial f}{\partial y}\cos\beta + \frac{\partial f}{\partial z}\cos\gamma\right)\cdot\cos\gamma$$

$$= \frac{\partial^2 f}{\partial x^2}\cos^2\alpha + \frac{\partial^2 f}{\partial x\partial y}\cos\beta\cos\alpha + \frac{\partial^2 f}{\partial x\partial z}\cos\gamma\cos\alpha + \frac{\partial^2 f}{\partial y\partial x}\cos\alpha\cos\beta + \frac{\partial^2 f}{\partial y^2}\cos^2\beta$$

$$+ \frac{\partial^2 f}{\partial y\partial z}\cos\gamma\cos\beta + \frac{\partial^2 f}{\partial z\partial x}\cos\alpha\cos\gamma + \frac{\partial^2 f}{\partial z\partial y}\cos\beta\cos\gamma + \frac{\partial^2 f}{\partial z^2}\cos^2\gamma.$$

$$= \frac{\partial^2 f}{\partial x^2}\cos^2\alpha + \frac{\partial^2 f}{\partial y^2}\cos^2\beta + \frac{\partial^2 f}{\partial z^2}\cos^2\gamma + 2\frac{\partial^2 f}{\partial x\partial y}\cos\alpha\cos\beta + 2\frac{\partial^2 f}{\partial y\partial z}\cos\beta\cos\gamma + 2\frac{\partial^2 f}{\partial z\partial x}\cos\gamma\cos\alpha.$$

7. 设 $u = f(x, y, z)$ 具有连续的一阶偏导函数，设三方向 $\vec{l_1}(\cos\alpha_1, \cos\beta_1, \cos\gamma_1)$，$\vec{l_2}(\cos\alpha_2, \cos\beta_2, \cos\gamma_2)$，$\vec{l_3}(\cos\alpha_3, \cos\beta_3, \cos\gamma_3)$ 两两相垂直，验证：

(1) $\left(\frac{\partial u}{\partial \vec{l_1}}\right)^2 + \left(\frac{\partial u}{\partial \vec{l_2}}\right)^2 + \left(\frac{\partial u}{\partial \vec{l_3}}\right)^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2$；

(2) $\frac{\partial^2 u}{\partial \vec{l_1}^2} + \frac{\partial^2 u}{\partial \vec{l_2}^2} + \frac{\partial^2 u}{\partial \vec{l_3}^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$.

证明: (1) $\frac{\partial u}{\partial \vec{l_i}} = \frac{\partial u}{\partial x}\cos\alpha_i + \frac{\partial u}{\partial y}\cos\beta_i + \frac{\partial u}{\partial z}\cos\gamma_i$. $i = 1, 2, 3$.

$$\left(\frac{\partial u}{\partial \vec{l_1}}\right)^2 + \left(\frac{\partial u}{\partial \vec{l_2}}\right)^2 + \left(\frac{\partial u}{\partial \vec{l_3}}\right)^2 = \left(\frac{\partial u}{\partial x}\right)^2(\cos^2\alpha_1 + \cos^2\alpha_2 + \cos^2\alpha_3)$$

$$+ \left(\frac{\partial u}{\partial y}\right)^2(\cos^2\beta_1 + \cos^2\beta_2 + \cos^2\beta_3) + \left(\frac{\partial u}{\partial z}\right)^2(\cos^2\gamma_1 + \cos^2\gamma_2 + \cos^2\gamma_3)$$

$$+ 2\frac{\partial u}{\partial x}\cdot\frac{\partial u}{\partial y}(\cos\alpha_1\cos\beta_1 + \cos\alpha_2\cos\beta_2 + \cos\alpha_3\cos\beta_3) + 2\frac{\partial u}{\partial y}\cdot\frac{\partial u}{\partial z}(\cos\beta_1\cos\gamma_1 + \cos\beta_2\cos\gamma_2 + \cos\beta_3\cos\gamma_3)$$

$$+ 2\frac{\partial u}{\partial z}\cdot\frac{\partial u}{\partial x}(\cos\gamma_1\cos\alpha_1 + \cos\gamma_2\cos\alpha_2 + \cos\gamma_3\cos\alpha_3).$$

$\because \vec{l_1} = (\cos\alpha_1, \cos\beta_1, \cos\gamma_1)$，$\vec{l_2} = (\cos\alpha_2, \cos\beta_2, \cos\gamma_2)$，$\vec{l_3} = (\cos\alpha_3, \cos\beta_3, \cos\gamma_3)$ 两两垂直，

$\therefore$ 有 $\cos^2\alpha_i + \cos^2\beta_i + \cos^2\gamma_i = 1$，$\cos\alpha_i\cos\alpha_j + \cos\beta_i\cos\beta_j + \cos\gamma_i\cos\gamma_j = 0$ $(i \neq j)$.

记 $A=\begin{pmatrix}\vec{l}_1\\ \vec{l}_2\\ \vec{l}_3\end{pmatrix}=\begin{pmatrix}\cos\alpha_1 & \cos\beta_1 & \cos\gamma_1\\ \cos\alpha_2 & \cos\beta_2 & \cos\gamma_2\\ \cos\alpha_3 & \cos\beta_3 & \cos\gamma_3\end{pmatrix}$ 则 $A^T=(\vec{l}_1^T\ \vec{l}_2^T\ \vec{l}_3^T)=\begin{pmatrix}\cos\alpha_1 & \cos\alpha_2 & \cos\alpha_3\\ \cos\beta_1 & \cos\beta_2 & \cos\beta_3\\ \cos\gamma_1 & \cos\gamma_2 & \cos\gamma_3\end{pmatrix}$

$$A\cdot A^T=\begin{pmatrix}\cos\alpha_1 & \cos\beta_1 & \cos\gamma_1\\ \cos\alpha_2 & \cos\beta_2 & \cos\gamma_2\\ \cos\alpha_3 & \cos\beta_3 & \cos\gamma_3\end{pmatrix}\begin{pmatrix}\cos\alpha_1 & \cos\alpha_2 & \cos\alpha_3\\ \cos\beta_1 & \cos\beta_2 & \cos\beta_3\\ \cos\gamma_1 & \cos\gamma_2 & \cos\gamma_3\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}=I.$$

即 A 为正交阵: $AA^T=A^TA=I$.

∴ 成立:

$$\begin{cases}\cos^2\alpha_1+\cos^2\alpha_2+\cos^2\alpha_3=\cos^2\alpha_1+\cos^2\beta_1+\cos^2\gamma_1=1.\\ \cos^2\beta_1+\cos^2\beta_2+\cos^2\beta_3=\cos^2\alpha_2+\cos^2\beta_2+\cos^2\gamma_2=1.\\ \cos^2\gamma_1+\cos^2\gamma_2+\cos^2\gamma_3=\cos^2\alpha_3+\cos^2\beta_3+\cos^2\gamma_3=1.\\ \cos\alpha_1\cos\beta_1+\cos\alpha_2\cos\beta_2+\cos\alpha_3\cos\beta_3=\cos\alpha_1\cos\alpha_2+\cos\beta_1\cos\beta_2+\cos\gamma_1\cos\gamma_2=0.\\ \cos\beta_1\cos\gamma_1+\cos\beta_2\cos\gamma_2+\cos\beta_3\cos\gamma_3=\cos\alpha_2\cos\alpha_3+\cos\beta_2\cos\beta_3+\cos\gamma_2\cos\gamma_3=0\\ \cos\gamma_1\cos\alpha_1+\cos\gamma_2\cos\alpha_2+\cos\gamma_3\cos\alpha_3=\cos\alpha_3\cos\alpha_1+\cos\beta_3\cos\beta_1+\cos\gamma_3\cos\gamma_1=0\end{cases}$$

$$\therefore\left(\frac{\partial u}{\partial \vec{l}_1}\right)^2+\left(\frac{\partial u}{\partial \vec{l}_2}\right)^2+\left(\frac{\partial u}{\partial \vec{l}_3}\right)^2=\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2+\left(\frac{\partial u}{\partial z}\right)^2$$

(2) 由第6题可知:

$$\frac{\partial^2 u}{\partial \vec{l}_1^{\,2}}+\frac{\partial^2 u}{\partial \vec{l}_2^{\,2}}+\frac{\partial^2 u}{\partial \vec{l}_3^{\,2}}=\frac{\partial^2 u}{\partial x^2}(\cos^2\alpha_1+\cos^2\alpha_2+\cos^2\alpha_3)+\frac{\partial^2 u}{\partial y^2}(\cos^2\beta_1+\cos^2\beta_2+\cos^2\beta_3)$$

$$+\frac{\partial^2 u}{\partial z^2}(\cos^2\gamma_1+\cos^2\gamma_2+\cos^2\gamma_3)+2\frac{\partial^2 u}{\partial x\partial y}(\cos\alpha_1\cos\beta_1+\cos\alpha_2\cos\beta_2+\cos\alpha_3\cos\beta_3)$$

$$+2\frac{\partial^2 u}{\partial y\partial z}(\cos\beta_1\cos\gamma_1+\cos\beta_2\cos\gamma_2+\cos\beta_3\cos\gamma_3)+2\frac{\partial^2 u}{\partial z\partial x}(\cos\gamma_1\cos\alpha_1+\cos\gamma_2\cos\alpha_2+\cos\gamma_3\cos\alpha_3)$$

$$=\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}.$$

§6 Taylor 公式

习题 P.58

1. 写出下列函数在原点处的二阶 Taylor 展开式.

(1) $z=e^{-x}\ln(1+y)$.

解: $\frac{\partial z}{\partial x}=-e^{-x}\ln(1+y)$. $\frac{\partial z}{\partial y}=\frac{e^{-x}}{1+y}$.

$\frac{\partial^2 z}{\partial x^2}=e^{-x}\ln(1+y)$. $\frac{\partial^2 z}{\partial x\partial y}=-\frac{e^{-x}}{1+y}$. $\frac{\partial^2 z}{\partial y^2}=-\frac{e^{-x}}{(1+y)^2}$.

$$\therefore z(x,y) = z(0,0) + \left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y}\right)z(0,0) + \frac{1}{2}\left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y}\right)^2 z(0,0) + R_2$$

$$= 0 + (x\cdot 0 + y\cdot 1) + \frac{1}{2}\left[x^2\cdot 0 + 2xy\cdot(-1) + y^2\cdot(-1)\right] + R_2$$

$$= y + \frac{1}{2}(-2xy - y^2) + R_2 = y - xy - \frac{1}{2}y^2 + R_2.$$

(2) $u = \ln(1+x+y+z)$

解: $\frac{\partial u}{\partial x} = \frac{1}{1+x+y+z} = \frac{\partial u}{\partial y} = \frac{\partial u}{\partial z}$.

$$\frac{\partial^2 u}{\partial x^2} = -\frac{1}{(1+x+y+z)^2} = \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial z^2} = \frac{\partial^2 u}{\partial x\partial y} = \frac{\partial^2 u}{\partial y\partial z} = \frac{\partial^2 u}{\partial z\partial x}$$

$$\therefore u(x,y,z) = u(0,0,0) + \left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y} + \Delta z\frac{\partial}{\partial z}\right)u(0,0,0) + \frac{1}{2}\left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y} + \Delta z\frac{\partial}{\partial z}\right)^2 u(0,0,0)$$

$$= 0 + (x\cdot 1 + y\cdot 1 + z\cdot 1) + \frac{1}{2}\left[x^2\cdot(-1) + y^2\cdot(-1) + z^2\cdot(-1) + 2xy\cdot(-1) + 2yz\cdot(-1) + 2zx\cdot(-1)\right] + R_2$$

$$= 0 + x + y + z - \frac{1}{2}(x^2 + y^2 + z^2 + 2xy + 2yz + 2zx) + R_2$$

$$= x + y + z - \frac{1}{2}x^2 - \frac{1}{2}y^2 - \frac{1}{2}z^2 - xy - yz - zx + R_2.$$

2. 求函数 $f(x,y) = \sin x\sin y$ 在点 $(\frac{\pi}{4}, \frac{\pi}{4})$ 处的二阶 Taylor 公式.

解: $\frac{\partial f}{\partial x} = \cos x\sin y$. $\frac{\partial f}{\partial y} = \sin x\cos y$.

$$\frac{\partial^2 f}{\partial x^2} = -\sin x\sin y. \quad \frac{\partial^2 f}{\partial x\partial y} = \cos x\cos y. \quad \frac{\partial^2 f}{\partial y^2} = -\sin x\sin y.$$

$$f(x,y) = f\left(\frac{\pi}{4},\frac{\pi}{4}\right) + \left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y}\right)f\left(\frac{\pi}{4},\frac{\pi}{4}\right) + \frac{1}{2}\left(\Delta x\frac{\partial}{\partial x} + \Delta y\frac{\partial}{\partial y}\right)^2 f\left(\frac{\pi}{4},\frac{\pi}{4}\right) + R_2$$

$$= \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}}{2} + \left[\left(x-\frac{\pi}{4}\right)\cdot\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}}{2} + \left(y-\frac{\pi}{4}\right)\cdot\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}}{2}\right] + \frac{1}{2}\left[\left(x-\frac{\pi}{4}\right)^2\cdot\left(-\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}}{2}\right)\right.$$

$$\left. + 2\left(x-\frac{\pi}{4}\right)\left(y-\frac{\pi}{4}\right)\cdot\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}}{2} + \left(y-\frac{\pi}{4}\right)^2\cdot\left(-\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}}{2}\right)\right] + R_2$$

$$= \frac{1}{2} + \frac{1}{2}\left(x-\frac{\pi}{4}\right) + \frac{1}{2}\left(y-\frac{\pi}{4}\right) - \frac{1}{4}\left(x-\frac{\pi}{4}\right)^2 + \frac{1}{2}\left(x-\frac{\pi}{4}\right)\left(y-\frac{\pi}{4}\right) - \frac{1}{4}\left(y-\frac{\pi}{4}\right)^2 + R_2$$

$$= \frac{1}{2} - \frac{\pi}{4} + \frac{1}{2}x + \frac{1}{2}y - \frac{1}{4}\left(x-\frac{\pi}{4}\right)^2 - \frac{1}{4}\left(y-\frac{\pi}{4}\right)^2 + \frac{1}{2}\left(x-\frac{\pi}{4}\right)\left(y-\frac{\pi}{4}\right) + R_2.$$

3. 求函数 $f(x,y)=e^{x+y}$ 在原点处的 n 阶 Taylor 公式.

解: $\frac{\partial f}{\partial x}=\frac{\partial f}{\partial y}=e^{x+y}$ $\quad \frac{\partial^{(n)} f}{\partial x^{(k)}\partial y^{(n-k)}}=e^{x+y}$

$$\therefore f(x,y)=f(0,0)+\sum_{k=1}^{n}\frac{1}{k!}\left(\Delta x\frac{\partial}{\partial x}+\Delta y\frac{\partial}{\partial y}\right)^{(k)}f(0,0)+R_n$$

$$=1+(x+y)+\frac{1}{2!}(x+y)^2+\cdots+\frac{1}{n!}(x+y)^n+R_n$$

$$R_n=\frac{1}{(n+1)!}(x+y)^{n+1}\cdot e^{\theta(x+y)} \qquad (0<\theta<1)$$

4. 利用 2 阶 Taylor 公式计算 $8.96^{2.03}$ 的近似值.

解: 设 $f(x,y)=(9+x)^{2+y}$ 在 $(0,0)$ 点的 2 阶 Taylor 展开.

$$f(x,y)=f(0,0)+\left(\Delta x\frac{\partial}{\partial x}+\Delta y\frac{\partial}{\partial y}\right)f(0,0)+\frac{1}{2}\left(\Delta x\frac{\partial}{\partial x}+\Delta y\frac{\partial}{\partial y}\right)^2 f(0,0)+R_2$$

$$=f(0,0)+x\left[(2+y)(9+x)^{1+y}\right]_{(0,0)}+y\left[\ln(9+x)\cdot(9+x)^{2+y}\right]_{(0,0)}$$

$$+\frac{1}{2}x^2\left[(2+y)(1+y)(9+x)^{y}\right]_{(0,0)}+xy\left[(9+x)^{1+y}+(2+y)\ln(9+x)\cdot(9+x)^{1+y}\right]_{(0,0)}$$

$$+\frac{1}{2}y^2\left[\ln^2(9+x)\cdot(9+x)^{2+y}\right]_{(0,0)}+R_2$$

$$=f(0,0)+18x+81\ln 9\cdot y+x^2+(9+18\ln 9)xy+\frac{81}{2}\ln^2 9\cdot y^2+R_2$$

$$\therefore 8.96^{2.03}=f(-0.04,-0.03)$$

$$\approx 81-18\times 0.04+81\times\ln 9\times 0.03+(-0.04)^2+(9+18\ln 9)(-0.04)(0.03)$$

$$+\frac{81}{2}\ln^2 9\cdot(-0.03)^2$$

$$=81-0.72+5.339+0.0016-0.0583+0.176=85.74$$

§7. 极值

习题. P.72.

1. 求函数 $f(x,y)=x^4+y^4-4xy+1$ 的极值

解: $\operatorname{grad} f=\begin{pmatrix}\frac{\partial f}{\partial x}\\ \frac{\partial f}{\partial y}\end{pmatrix}=\begin{pmatrix}4x^3-4y\\ 4y^3-4x\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\Rightarrow$ 驻点: $\vec{x}_1=\begin{pmatrix}-1\\-1\end{pmatrix}$ $\vec{x}_2=\begin{pmatrix}0\\0\end{pmatrix}$ $\vec{x}_3=\begin{pmatrix}1\\1\end{pmatrix}$

$$\nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{bmatrix}.$$

$\nabla^2 f(\vec{x}_1) = \begin{bmatrix} 12 & -4 \\ -4 & 12 \end{bmatrix}$. 正定. $\therefore \vec{x}_1 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$ 为极小值点. $f(\vec{x}_1) = -1$.

$\nabla^2 f(\vec{x}_2) = \begin{bmatrix} 0 & -4 \\ -4 & 0 \end{bmatrix}$. 不定. $\Delta = -16 < 0$. $\therefore \vec{x}_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ 非极值点.

$\nabla^2 f(\vec{x}_3) = \begin{bmatrix} 12 & -4 \\ -4 & 12 \end{bmatrix}$. 正定. $\therefore \vec{x}_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ 为极小值点. $f(\vec{x}_3) = -1$.

2. 求函数 $f(x,y) = e^{2x}(x + y^2 + 2y)$ 的极值.

解: $\nabla f = \begin{pmatrix} f'_x \\ f'_y \end{pmatrix} = \begin{pmatrix} e^{2x}(2x + 2y^2 + 4y + 1) \\ e^{2x}(2y + 2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow$ 驻点: $\vec{x}_1 = \begin{pmatrix} \frac{1}{2} \\ -1 \end{pmatrix}$

$\nabla^2 f = \begin{bmatrix} f''_{xx} & f''_{xy} \\ f''_{xy} & f''_{yy} \end{bmatrix} = \begin{bmatrix} 4e^{2x}(x + y^2 + 2y + 1) & 4e^{2x}(y+1) \\ 4e^{2x}(y+1) & 2e^{2x} \end{bmatrix}$. $\nabla^2 f(\vec{x}_1) = \begin{bmatrix} 2e & 0 \\ 0 & 2e \end{bmatrix}$ 正定.

$\therefore \vec{x}_1 = \begin{pmatrix} \frac{1}{2} \\ -1 \end{pmatrix}$ 为极小值点. $f_{min} = -\frac{e}{2}$.

3. 讨论 $f(x,y) = y^2 - x^2$ 的极值.

解: $\nabla f = \begin{pmatrix} f'_x \\ f'_y \end{pmatrix} = \begin{pmatrix} -2x \\ 2y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow$ 驻点: $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

$\nabla^2 f = \begin{bmatrix} f''_{xx} & f''_{xy} \\ f''_{xy} & f''_{yy} \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & 2 \end{bmatrix}$ 不定. $\therefore \vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ 非极值点. $f(x,y)$ 无极值.

4. 讨论函数 $f(x,y) = (y - x^2)(y - x^4)$ 的极值.

解: $\nabla f = \begin{pmatrix} f'_x \\ f'_y \end{pmatrix} = \begin{pmatrix} 6x^5 - 4x^3 y - 2xy \\ -x^4 - x^2 + 2y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow$ 驻点:

$\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. $\vec{x}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. $\vec{x}_3 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. $\vec{x}_4 = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{3}{8} \end{pmatrix}$. $\vec{x}_5 = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{3}{8} \end{pmatrix}$.

$\nabla^2 f = \begin{bmatrix} f''_{xx} & f''_{xy} \\ f''_{xy} & f''_{yy} \end{bmatrix} = \begin{bmatrix} 30x^4 - 12x^2 y - 2y & -4x^3 - 2x \\ -4x^3 - 2y & 2 \end{bmatrix}$

$\Delta = 2(30x^4 - 12x^2 y - 2y) - (4x^3 + 2y)^2 = 4(15x^4 - 6x^2 y - y) - 4(2x^3 + y)^2$.

$\Delta(\vec{x}_1) = 0$. 且 $f(\vec{x}_1) = f(0,0) = 0$. $f(x, x^3) = -x^5(1 - x^2)$ 在 $(0,0)$ 点附近变号.

$\therefore \vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ 非极值点.

$\Delta(\vec{x}_2) = -20 < 0$. $\Delta(\vec{x}_3) = -20 < 0$. $\therefore \vec{x}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ 和 $\vec{x}_3 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ 均非极值点.

$\Delta(\vec{x}_4) = 1 > 0$. $f''_{xx}(\vec{x}_4) = \frac{9}{2} > 0$.

$\Delta(\vec{x}_5) = 1 > 0$. $f''_{xx}(\vec{x}_5) = \frac{9}{2} > 0$.

$\therefore \vec{x}_{4,5} = \begin{pmatrix} \pm\frac{\sqrt{2}}{2} \\ \frac{3}{8} \end{pmatrix}$ 为极小值点. $f_{min} = -\frac{1}{64}$.

5. 证明函数 $f(x,y) = (1+e^y)\cos x - ye^y$ 有无穷多个极大值点，但无极小值点.

证明: $\nabla f = \begin{pmatrix} f'_x \\ f'_y \end{pmatrix} = \begin{pmatrix} -(1+e^y)\sin x \\ e^y(\cos x - 1 - y) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow$ 驻点: $\vec{x}_k = \begin{pmatrix} k\pi \\ (-1)^k - 1 \end{pmatrix}$. $k = 0, \pm 1, \pm 2, \cdots$

$\nabla^2 f = \begin{bmatrix} f''_{xx} & f''_{xy} \\ f''_{xy} & f''_{yy} \end{bmatrix} = \begin{bmatrix} -(1+e^y)\cos x & -e^y \sin x \\ -e^y \sin x & e^y(\cos x - y - 2) \end{bmatrix}$.

$\nabla^2 f(\vec{x}_k) = \begin{bmatrix} (-1)^{k+1}(1+e^{(-1)^k - 1}) & 0 \\ 0 & -e^{(-1)^k - 1} \end{bmatrix}$

当 $k = 2n$ 时. $\nabla^2 f(\vec{x}_{2n}) = \begin{pmatrix} -2 & 0 \\ 0 & -1 \end{pmatrix}$ 负定. $\therefore \vec{x}_{2n} = \begin{pmatrix} 2n\pi \\ 0 \end{pmatrix}$ 为极大值点，有无穷多个.

当 $k = 2n+1$ 时. $\nabla^2 f(\vec{x}_{2n+1}) = \begin{pmatrix} 1+\frac{1}{e^2} & 0 \\ 0 & -\frac{1}{e^2} \end{pmatrix}$ 不定. $\therefore \vec{x}_{2n+1} = \begin{pmatrix} (2n+1)\pi \\ 0 \end{pmatrix}$ 非极值点.

6. 函数 $z = z(x,y)$ 由下列方程确定，试求其极值:

(1) $2x^2 + 2y^2 + z^2 + 8xz - z + 8 = 0$.

解: $\nabla f = \begin{pmatrix} f'_x \\ f'_y \end{pmatrix} = -\frac{1}{F'_z}\begin{pmatrix} F'_x \\ F'_y \end{pmatrix} = -\frac{1}{2z+8x-1}\begin{pmatrix} 4x+8z \\ 4y \end{pmatrix} = 0 \Rightarrow$ 驻点: $\vec{x}_1 = \begin{pmatrix} \frac{16}{7} \\ 0 \\ -\frac{8}{7} \end{pmatrix}$. $\vec{x}_2 = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$

$f''_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left(-\frac{4x+8z}{2z+8x-1}\right) = \frac{-(4+8f'_x)(2z+8x-1) + (4x+8z)(2f'_x+8)}{(2z+8x-1)^2}$

$f''_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}\left(-\frac{4x+8z}{2z+8x-1}\right) = \frac{-8f'_y(2z+8x-1) + (4x+8z)\cdot 2f'_y}{(2z+8x-1)^2}$

$= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(-\frac{4y}{2z+8x-1}\right) = \frac{4y(2f'_x+8)}{(2z+8x-1)^2}$

$f''_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(-\frac{4y}{2z+8x-1}\right) = \frac{-4(2z+8x-1) + 4y(2f'_y)}{(2z+8x-1)^2}$

在驻点处: $f'_x = 0$. $f'_y = 0$ 即: $4x+8z = 0$. $y = 0$. ($\vec{x}_2$ 舍去)

即: $f''_{xx}(\vec{x}_1) = \frac{-4(2z+8x-1) + 8(4x+8z)}{(2z+8x-1)^2} = -\frac{4}{2z+8x-1} = \frac{4}{1-7x}$.

$f''_{xy}(\vec{x}_i)=0$. $\quad f''_{yy}(\vec{x}_i)=\dfrac{-4}{2z+8x-1}=\dfrac{4}{1-7x}$.

$\Delta(\vec{x}_i)=(f''_{xx}f''_{yy}-f''^{2}_{xy})(\vec{x}_i)=\left(\dfrac{4}{1-7x}\right)^2>0$.

$\because f''_{xx}(\vec{x}_1)=\dfrac{4}{1-7x_1}=\dfrac{4}{1-7\cdot\frac{16}{5}}=-\dfrac{4}{15}<0$. $\therefore \vec{x}_1=\begin{pmatrix}\frac{16}{5}\\0\\-\frac{8}{5}\end{pmatrix}$ 为极大值点. $f_{max}=-\dfrac{8}{5}$.

$f''_{xx}(\vec{x}_2)=\dfrac{4}{1-7x_2}=\dfrac{4}{1-7(-2)}=\dfrac{4}{15}>0$. $\therefore \vec{x}_2=\begin{pmatrix}-2\\0\\1\end{pmatrix}$ 为极小值点. $f_{min}=1$.

(2) $x^2+\dfrac{y^2}{4}+\dfrac{z^2}{9}-\dfrac{2}{3}z=0$.

解: $\nabla f=\begin{pmatrix}z'_x\\z'_y\end{pmatrix}=-\dfrac{1}{F_z}\begin{pmatrix}F_x\\F_y\end{pmatrix}=-\dfrac{1}{\frac{2}{9}z-\frac{2}{3}}\begin{pmatrix}2x\\\frac{1}{2}y\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\Rightarrow$ 驻点: $\vec{x}_1=\begin{pmatrix}0\\0\\0\end{pmatrix}$. $\vec{x}_2=\begin{pmatrix}0\\0\\6\end{pmatrix}$.

$z''_{xx}=\dfrac{\partial}{\partial x}(z'_x)=\dfrac{\partial}{\partial x}\left(\dfrac{9x}{3-z}\right)=\dfrac{9(3-z)+9x\cdot z'_x}{(3-z)^2}$. $\quad z''_{xx}(\vec{x}_i)=\dfrac{9}{3-z}$.

$z''_{xy}=\dfrac{\partial}{\partial y}(z'_x)=\dfrac{\partial}{\partial y}\left(\dfrac{9x}{3-z}\right)=\dfrac{9x}{(3-z)^2}\cdot z'_y$. $\quad z''_{xy}(\vec{x}_i)=0$.

$z''_{yy}=\dfrac{\partial}{\partial y}(z'_y)=\dfrac{\partial}{\partial y}\left(\dfrac{9y}{4(3-z)}\right)=\dfrac{9}{4}\cdot\dfrac{3-z+y\cdot 2z'_y}{(3-z)^2}$. $\quad z''_{yy}(\vec{x}_i)=\dfrac{9}{4(3-z)}$.

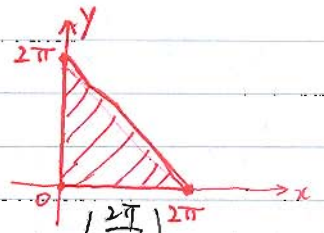
$\Delta(\vec{x}_i)=z''_{xx}z''_{yy}-z''^{2}_{xy}=\dfrac{81}{4(3-z)^2}>0$. $\quad z''_{xx}(\vec{x}_1)=3>0$. $\quad z''_{xx}(\vec{x}_2)=-3<0$.

$\therefore \vec{x}_1=\begin{pmatrix}0\\0\\0\end{pmatrix}$ 为极小值点. $z_{min}=0$. $\quad \vec{x}_2=\begin{pmatrix}0\\0\\6\end{pmatrix}$ 为极大值点. $z_{max}=6$.

7. 求函数 $f(x,y)=\sin x+\sin y-\sin(x+y)$ 在闭区域

$D=\{(x,y)\mid x\geq 0,\ y\geq 0,\ x+y\leq 2\pi\}$

上的最大值和最小值.



解: $\nabla f=\begin{pmatrix}f'_x\\f'_y\end{pmatrix}=\begin{pmatrix}\cos x-\cos(x+y)\\\cos y-\cos(x+y)\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\Rightarrow$ 驻点: $\vec{x}_1=\begin{pmatrix}0\\0\end{pmatrix}$. $\vec{x}_2=\begin{pmatrix}\frac{2\pi}{3}\\\frac{2\pi}{3}\end{pmatrix}$. $\vec{x}_1,\vec{x}_2\in D$.

$f''_{xx}=-\sin x+\sin(x+y)$. $\quad f''_{xx}(\vec{x}_1)=0$. $\quad f''_{xx}(\vec{x}_2)=-\sin\frac{2\pi}{3}+\sin\frac{4\pi}{3}=-1$.

$f''_{xy}=\sin(x+y)$. $\quad f''_{xy}(\vec{x}_1)=0$. $\quad f''_{xy}(\vec{x}_2)=\sin\frac{4\pi}{3}=-\frac{1}{2}$.

$f''_{yy} = -\sin y + \sin(x+y)$ $\quad f''_{yy}(\vec{x}_1) = 0$. $\quad f''_{yy}(\vec{x}_2) = -\sin\frac{\pi}{3} + \sin\frac{4\pi}{3} = -1$.

$\Delta(\vec{x}_1) = 0$. $\quad \Delta(\vec{x}_2) = 1 - \frac{1}{4} = \frac{3}{4} > 0$. $\quad f''_{xx}(\vec{x}_2) = -1 < 0$.

$\therefore \vec{x}_2 = \begin{pmatrix} \frac{\pi}{3} \\ \frac{2\pi}{3} \end{pmatrix}$ 是极大值点. $f_{max} = \frac{3\sqrt{3}}{2}$.

显然, $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ 是区域D边界上的点. $f(\vec{x}_1) = 0$.

在边界 $x=0$, $y=0$, $x+y=2\pi$ 上都有 $f(x,y)=0$. 而 $\vec{x}_1$ 是区域D内唯一的驻点.

$\therefore$ 最大值 $f_{max} = \frac{3\sqrt{3}}{2}$. 最小值 $f_{min} = 0$

8. 求 $f(x,y) = (ax^2 + by^2)e^{-x^2-y^2}$ 的最大值与最小值 $(a \neq b)$.

解: $\left.\begin{aligned} f'_x &= 2ax \cdot e^{-x^2-y^2} - 2x(ax^2+by^2)e^{-x^2-y^2} = 2xe^{-x^2-y^2}(a - ax^2 - by^2) = 0 \\ f'_y &= 2by \cdot e^{-x^2-y^2} - 2y(ax^2+by^2)e^{-x^2-y^2} = 2ye^{-x^2-y^2}(b - ax^2 - by^2) = 0 \end{aligned}\right\} \Rightarrow$

驻点: $\vec{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. $\quad \vec{x}_{2,3} = \begin{pmatrix} 0 \\ \pm 1 \end{pmatrix}$. $\quad \vec{x}_{4,5} = \begin{pmatrix} \pm 1 \\ 0 \end{pmatrix}$.

$f(\vec{x}_1) = 0$. $\quad f(\vec{x}_{2,3}) = \frac{b}{e}$. $\quad f(\vec{x}_{4,5}) = \frac{a}{e}$.

$\because f(x,y)$ 在全平面上连续有界. $\therefore$ 必能取到最大、最小值.

$f_{max} = \max\left(0, \frac{b}{e}, \frac{a}{e}\right)$. $\quad f_{min} = \min\left(0, \frac{b}{e}, \frac{a}{e}\right)$.

9. 证明: 当 $\sqrt{x^2+y^2} < 1$ 时, 成立不等式

$$2|xy| - \frac{x^2y^2}{6} \leq 4 - 4\cos\sqrt{|xy|} \leq 2|xy|$$

证明: 将 $\cos z$ 在 0 点邻近作 Taylor 展开.

$$\cos z = 1 - \frac{1}{2}z^2 + \frac{1}{24}z^4 - \frac{1}{6!}z^6 + \frac{1}{8!}z^8 - \cdots$$

当 $0 \leq z < 1$ 时, 显然有 $1 - \frac{1}{2}z^2 \leq \cos z \leq 1 - \frac{1}{2}z^2 + \frac{1}{24}z^4$.

当 $\sqrt{x^2+y^2} < 1$ 时, $\sqrt{|xy|} \leq \sqrt{\frac{x^2+y^2}{2}} < \frac{\sqrt{2}}{2} < 1$. 令 $z = \sqrt{|xy|}$. 代入上式有:

$$1 - \frac{|xy|}{2} \leq \cos\sqrt{|xy|} \leq 1 - \frac{|xy|}{2} + \frac{x^2y^2}{24}$$

即: $2|xy| - \frac{x^2y^2}{6} \leq 4 - 4\cos\sqrt{|xy|} \leq 2|xy|$.

10. 某养殖场饲养两种鱼，若甲种鱼放养 x（万尾），乙种鱼放养 y（万尾），收获时两种鱼的收获量分别为 $(3-\alpha x-\beta y)x$ 和 $(4-\beta x-2\alpha y)y$ $(\alpha>\beta>0)$.
求使产鱼总量最大的放养数.

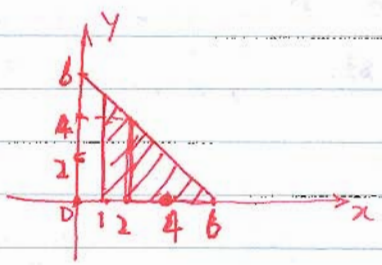
解: $f(x,y)=(3-\alpha x-\beta y)x+(4-\beta x-2\alpha y)y$

$$\left.\begin{aligned} f'_x &= 3-\alpha x-\beta y-\alpha x-\beta y = 3-2\alpha x-2\beta y=0 \\ f'_y &= -\beta x+4-\beta x-2\alpha y-2\alpha y = 4-2\beta x-4\alpha y=0 \end{aligned}\right\} \Rightarrow \text{驻点: } \begin{cases} x=\dfrac{2\alpha-2\beta}{2\alpha^2-\beta^2} \\ y=\dfrac{4\alpha-3\beta}{2(2\alpha^2-\beta^2)} \end{cases}$$

$f''_{xx}=2\alpha$. $f''_{xy}=-2\beta$. $f''_{yy}=-4\alpha$.

$\Delta=8\alpha^2-4\beta^2>0$. $(\alpha>\beta>0)$ $f''_{xx}=-2\alpha<0$. ∴ 驻点为极大值点.

11. 求 $f(x,y)=xy(4-x-y)$ 在 $x=1$, $y=0$, $x+y=6$ 所围区域上的最大值与最小值.



解: $\left.\begin{aligned} f'_x &= 4y-2xy-y^2=0 \\ f'_y &= 4x-x^2-2xy=0 \end{aligned}\right\}$ 驻点:

$\vec{x}_1=\begin{pmatrix}0\\0\end{pmatrix}$ 为在区域内. $\vec{x}_2=\begin{pmatrix}4\\0\end{pmatrix}$ 边界点. $f(\vec{x}_1)=0$.

$\vec{x}_3=\begin{pmatrix}0\\4\end{pmatrix}$ 为在区域内. $\vec{x}_4=\begin{pmatrix}\frac{4}{3}\\\frac{4}{3}\end{pmatrix}$ 区域内点. $f(\vec{x}_4)=\dfrac{64}{27}$.

$y=0$ 上边界点 $f=0$.

$x=1$ 上边界点 $-10\le f\le\dfrac{9}{4}$ $(f(x,y)=y(3-y))$.

$x+y=6$ 上边界点 $-18\le f\le 0$. $(f(x,y)=-2y(6-y))$

∴ $f_{max}=f\left(\frac{4}{3},\frac{4}{3}\right)=\dfrac{64}{27}$. $f_{min}=f(3,3)=-18$.

12. 要做一个体积为 $2m^3$ 的有盖的长方体容器，当长、宽、高取怎样的尺寸时，才能使用料最省.

解: 设长 x，宽 y，高 z，则 $V=xyz=2$. $S=2(xy+yz+zx)=S(x,y,z)$.

$$\begin{cases} \min & 2(xy+yz+zx) \\ s.t & xyz=2. \end{cases}$$

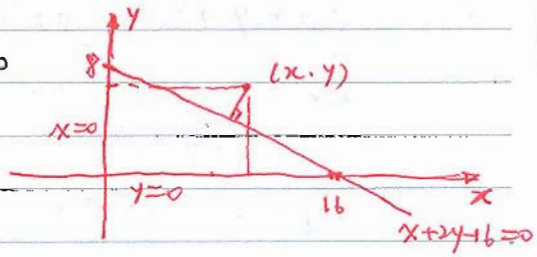
$L(x,y,z,\lambda)=2(xy+yz+zx)+\lambda(xyz-2)$.

$$\begin{cases} L'_x = 2(y+z) - \lambda yz = 0 \\ L'_y = 2(x+z) - \lambda xz = 0 \\ L'_z = 2(x+y) - \lambda xy = 0 \\ L'_\lambda = xyz - 2 = 0. \end{cases} \Rightarrow x=y=z=\sqrt[3]{2}.$$

即正方形时用料最省.

(由物理意义可知其最小值存在, 驻点唯一)

13. 在平面 Oxy 上求一点, 使它到 $x=0$, $y=0$ 及 $x+2y-16=0$ 三直线的距离平方和为最小.



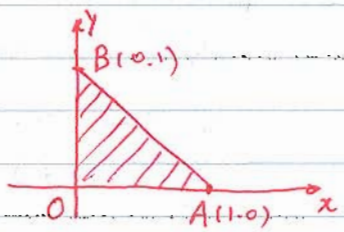
解: $d(x,y) = x^2+y^2+\frac{(x+2y-16)^2}{5}$

$$\begin{cases} d'_x = 2x + \frac{2}{5}(x+2y-16) = \frac{12}{5}x + \frac{4}{5}y - \frac{32}{5} = 0 \\ d'_y = 2y + \frac{4}{5}(x+2y-16) = \frac{4}{5}x + \frac{18}{5}y - \frac{64}{5} = 0 \end{cases} \Rightarrow \text{驻点} \begin{cases} x = \frac{8}{5} \\ y = \frac{16}{5}. \end{cases}$$

由问题意义可知必存在最小值, 且域内驻点唯一.

$$\therefore d_{min} = d\left(\frac{8}{5}, \frac{16}{5}\right) = \frac{128}{5}.$$

14. 在以 $O(0,0)$, $A(1,0)$, $B(0,1)$ 为顶点的三角形所围成的闭区域上找点, 使它到三个顶点的距离平方和取最大和最小.



解: $d(x,y) = x^2+y^2+(x-1)^2+y^2+x^2+(y-1)^2$

$= 3x^2+3y^2-2x-2y+2.$

$$\begin{cases} d'_x = 6x-2 = 0 \\ d'_y = 6y-2 = 0 \end{cases} \Rightarrow \text{驻点} \begin{cases} x = \frac{1}{3} \\ y = \frac{1}{3} \end{cases} \quad d\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{4}{3}.$$

$x=0$ 边界点: $d(0,y) = 3y^2-2y+2 = 3\left(y-\frac{1}{3}\right)^2+\frac{5}{3}$. $\frac{5}{3} \le d(0,y) \le 3$.

$y=0$ 边界点: $d(x,0) = 3x^2-2x+2 = 3\left(x-\frac{1}{3}\right)^2+\frac{5}{3}$. $\frac{5}{3} \le d(x,0) \le 3$.

$x+y=1$ 边界点: $d(x,1-x) = 3x^2+3(1-x)^2 = 6\left(x-\frac{1}{2}\right)^2+\frac{3}{2}$. $\frac{3}{2} \le d(x,1-x) \le 3$.

$$\therefore d_{min} = d\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{4}{3}. \quad d_{max} = d(1,0) = d(0,1) = 3.$$

15. 已知 n 个点 $P_i(a_i, b_i)$ $(i=1,2,\cdots,n)$, 求点 $P(x,y)$, 使其到 $P_1, P_2, \cdots, P_n$ 的距离平方和最小.

解: $d(x,y) = \sum_{i=1}^{n}\left((x-a_i)^2+(y-b_i)^2\right)$

$$d'_x = 2\sum_{i=1}^{n}(x-a_i) = 2nx - 2\sum_{i=1}^{n}a_i = 0$$

$$d'_y = 2\sum_{i=1}^{n}(y-b_i) = 2ny - 2\sum_{i=1}^{n}b_i = 0.$$

$\Rightarrow$ 驻点: $\begin{cases} x=\frac{1}{n}\sum_{i=1}^{n}a_i \\ y=\frac{1}{n}\sum_{i=1}^{n}b_i \end{cases}$ 由问题意义可知, 在驻点处达到 d_{min}.

16. 已知 $u=ax^2+by^2+cz^2$, 其中 a、b、c 均为正数, 求在约束条件 $x+y+z=1$ 下 u 的最小值.

解: $L(x,y,z,\lambda)=ax^2+by^2+cz^2+\lambda(x+y+z-1)$

$$\left.\begin{aligned} L'_x&=2ax+\lambda=0 \\ L'_y&=2by+\lambda=0 \\ L'_z&=2cz+\lambda=0 \\ L'_\lambda&=x+y+z-1=0 \end{aligned}\right\} \Rightarrow \begin{cases} x=\dfrac{bc}{ab+bc+ca} \\ y=\dfrac{ca}{ab+bc+ca} \\ z=\dfrac{ab}{ab+bc+ca} \end{cases}$$

由问题意义可知该点达到最小值.

$u_{min}=\dfrac{abc}{ab+bc+ca}$.

17. 求原点到 $z^2=xy+x-y+4$ 的最短距离.

解: $d^2(x,y,z)=x^2+y^2+z^2$.

$\min\ d(x,y,z)$

s.t $z^2=xy+x-y+4$

$L(x,y,z,\lambda)=x^2+y^2+z^2+\lambda(z^2-xy-x+y-4)$

$$\left.\begin{aligned} L'_x&=2x-\lambda y-\lambda=0 \\ L'_y&=2y-\lambda x+\lambda=0 \\ L'_z&=2z+2\lambda z=0 \\ L'_\lambda&=z^2-xy-x+y-4=0 \end{aligned}\right\} \Rightarrow \vec{x_1}=\begin{pmatrix}-1\\1\\\pm1\end{pmatrix},\ \vec{x_2}=\begin{pmatrix}1+\sqrt{5}\\-1-\sqrt{5}\\0\end{pmatrix},\ \vec{x_3}=\begin{pmatrix}1-\sqrt{5}\\-1+\sqrt{5}\\0\end{pmatrix}$$

$d^2(\vec{x_1})=3$, $d^2(\vec{x_2})=12+4\sqrt{5}$, $d^2(\vec{x_3})=12-4\sqrt{5}$.

$\therefore d^2_{min}=d(\vec{x_1})=d(-1,1,\pm1)=3$, $d_{min}=\sqrt{3}$.

18. 抛物面 $z=x^2+y^2$ 被平面 $x+y+z=1$ 截得一个椭圆, 求原点到这个椭圆上点的最长和最短距离.

解: $\min\ d^2(x,y,z)=x^2+y^2+z^2$

s.t $\begin{cases} z=x^2+y^2 \\ x+y+z=1 \end{cases}$

$L(x,y,z,\lambda,\mu)=x^2+y^2+z^2+\lambda(z-x^2-y^2)+\mu(x+y+z-1)$

$$\left.\begin{aligned} L'_x&=2x-2\lambda x+\mu=0 \\ L'_y&=2y-2\lambda y+\mu=0 \\ L'_z&=2z+\lambda+\mu=0 \\ L'_\lambda&=z-x^2-y^2=0 \\ L'_\mu&=x+y+z-1=0 \end{aligned}\right\} \Rightarrow \begin{cases} x=y=\dfrac{\mu}{2(\lambda-1)} \\ z=-\dfrac{\lambda+\mu}{2} \\ z=x^2+y^2 \\ x+y+z=1 \end{cases} \Rightarrow \vec{x_1}=\begin{pmatrix}\frac{-1+\sqrt{3}}{2}\\\frac{-1+\sqrt{3}}{2}\\2-\sqrt{3}\end{pmatrix},\ \vec{x_2}=\begin{pmatrix}\frac{-1-\sqrt{3}}{2}\\\frac{-1-\sqrt{3}}{2}\\2+\sqrt{3}\end{pmatrix}$$

$d^2(\vec{x}_1) = 9-5\sqrt{3}$. $d^2(\vec{x}_2) = 9+5\sqrt{3}$.

由问题意义可知: $d_{min} = d(\vec{x}_1) = \sqrt{9-5\sqrt{3}}$.

$d_{max} = d(\vec{x}_2) = \sqrt{9+5\sqrt{3}}$.

19. 求旋转抛物面 $z = x^2+y^2$ 与平面 $x+y-z=1$ 之间的最短距离.

解: $\min \quad d^2(x,y,z) = \frac{(x+y-z-1)^2}{3}$

s.t $\quad z = x^2+y^2$

$L(x,y,z,\lambda) = \frac{1}{3}(x+y-z-1)^2 + \lambda(z-x^2-y^2)$

$$\left.\begin{aligned} L'_x &= \tfrac{2}{3}(x+y-z-1) - 2\lambda x = 0 \\ L'_y &= \tfrac{2}{3}(x+y-z-1) - 2\lambda y = 0 \\ L'_z &= -\tfrac{2}{3}(x+y-z-1) + \lambda = 0 \\ L'_\lambda &= z - x^2 - y^2 = 0. \end{aligned}\right\} \Rightarrow \vec{x} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

由问题意义可知: $d_{min} = d(\vec{x}) = \frac{\sqrt{3}}{6}$.

20. 求函数 $f(x,y,z) = x^2+2y^2+z^2-2xy-2yz$ 在条件 $x^2+y^2+z^2=4$ 下的最大值与最小值.

解: $L(x,y,z,\lambda) = x^2+2y^2+z^2-2xy-2yz+\lambda(x^2+y^2+z^2-4)$

$$\left.\begin{aligned} L'_x &= 2x-2y+2\lambda x = 0. \\ L'_y &= 4y-2x-2z+2\lambda y = 0. \\ L'_z &= 2z-2y+2\lambda z = 0. \\ L'_\lambda &= x^2+y^2+z^2-4 = 0. \end{aligned}\right\} \Rightarrow \vec{x}_1 = \begin{pmatrix} \pm\sqrt{2} \\ 0 \\ \mp\sqrt{2} \end{pmatrix} \quad \vec{x}_2 = \begin{pmatrix} \pm\frac{2\sqrt{3}}{3} \\ \pm\frac{2\sqrt{3}}{3} \\ \pm\frac{2\sqrt{3}}{3} \end{pmatrix} \quad \vec{x}_3 = \begin{pmatrix} \pm\frac{\sqrt{6}}{3} \\ -\frac{2\sqrt{6}}{3} \\ \pm\frac{\sqrt{6}}{3} \end{pmatrix}$$

$f(\vec{x}_1) = 4$. $f(\vec{x}_2) = 0$. $f(\vec{x}_3) = 12$. $\therefore f_{max} = f(\vec{x}_3) = 12$.

$f_{min} = f(\vec{x}_2) = 0$.

21. 在上半椭球体 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2} \le 1$, $z \ge 0$ 内嵌入一个体积最大的长方体, 求其长、宽、高及体积.

解: 设长 x, 宽 y, 高 z, 则: $\max \quad V(x,y,z) = xyz$.

(由对称性知: $(\frac{x}{2}, \frac{y}{2}, z) \in$ 椭球面) s.t $\frac{x^2}{4a^2}+\frac{y^2}{4b^2}+\frac{z^2}{c^2} = 1$.

$L(x,y,z,\lambda) = xyz + \lambda\left(\frac{x^2}{4a^2}+\frac{y^2}{4b^2}+\frac{z^2}{c^2}-1\right)$.

$$\left.\begin{aligned} L'_x &= yz + \frac{1}{2a^2}x\lambda = 0 \\ L'_y &= zx + \frac{1}{2b^2}y\lambda = 0 \\ L'_z &= xy + \frac{1}{c^2}z\lambda = 0 \\ L'_\lambda &= \frac{x^2}{4a^2} + \frac{y^2}{4b^2} + \frac{z^2}{c^2} - 1 = 0. \end{aligned}\right\} \Rightarrow \vec{x} = \begin{pmatrix} \frac{2\sqrt{3}}{3}a \\ \frac{2\sqrt{3}}{3}b \\ \frac{\sqrt{3}}{3}c \end{pmatrix}$$

由题意可知，该点为最大值点。

此时长为 $\frac{2\sqrt{3}}{3}a$，宽为 $\frac{2\sqrt{3}}{3}b$，高为 $\frac{\sqrt{3}}{3}c$ 时，$V_{max} = \frac{4\sqrt{3}}{9}abc$.

22. 求 $z = \frac{1}{2}(x^4+y^4)$ 在条件 $x+y=a$ 下的最小值，其中 $x\geq 0$，$y\geq 0$，a 为常数，并证明不等式：$\frac{x^4+y^4}{2} \geq \left(\frac{x+y}{2}\right)^4$.

解：$L(x,y,\lambda) = \frac{1}{2}(x^4+y^4) + \lambda(x+y-a)$.

$$\left.\begin{aligned} L'_x &= 2x^3 + \lambda = 0 \\ L'_y &= 2y^3 + \lambda = 0 \\ L'_\lambda &= x+y-a = 0. \end{aligned}\right\} \Rightarrow x = y = \frac{a}{2}, \quad z\left(\frac{a}{2},\frac{a}{2}\right) = \frac{a^4}{16}.$$

由题意可知：$z_{min} = z\left(\frac{a}{2},\frac{a}{2}\right) = \frac{a^4}{16}$.

$\therefore z = \frac{1}{2}(x^4+y^4) \geq z_{min} = \frac{a^4}{16} = \left(\frac{a}{2}\right)^4 = \left(\frac{x+y}{2}\right)^4$.

即：$\frac{x^4+y^4}{2} \geq \left(\frac{x+y}{2}\right)^4$.

23. 当 $x>0$，$y>0$，$z>0$ 时，求函数 $f(x,y,z) = \ln x + 2\ln y + 3\ln z$ 在球面 $x^2+y^2+z^2 = 6R^2$ 上的最大值，并由此证明：当 a、b、c 为正实数时，成立不等式：$ab^2c^3 \leq 108\left(\frac{a+b+c}{6}\right)^6$.

解：$L(x,y,z,\lambda) = \ln x + 2\ln y + 3\ln z + \lambda(x^2+y^2+z^2-6R^2)$ $(x>0,\ y>0,\ z>0)$.

$$\left.\begin{aligned} L'_x &= \frac{1}{x} + 2\lambda x = 0 \\ L'_y &= \frac{2}{y} + 2\lambda y = 0 \\ L'_z &= \frac{3}{z} + 2\lambda z = 0 \\ L'_\lambda &= x^2+y^2+z^2-6R^2 = 0. \end{aligned}\right\} \Rightarrow \vec{x} = \begin{pmatrix} R \\ \sqrt{2}R \\ \sqrt{3}R \end{pmatrix}$$

此时有：$f_{max} = f(\vec{x}) = \ln(6\sqrt{3}R^6)$.

令 $x^2=a$，$y^2=b$，$z^2=c$，则有：

$$\ln xy^2z^3 \leq \ln(6\sqrt{3}R^6) = \ln\left[6\sqrt{3}\cdot\left(\frac{x^2+y^2+z^2}{6}\right)^3\right]$$

即：$\ln ab^2c^3 \leq \ln\left[6\sqrt{3}\cdot\left(\frac{x^2+y^2+z^2}{6}\right)\right]^2 = \ln 108\cdot\left(\frac{a+b+c}{6}\right)^6$

$\therefore ab^2c^3 \leq 108\left(\frac{a+b+c}{6}\right)^6$.

24. 某时间通过电台和报纸两种方式作销售某种商品广告，根据统计资料，销售收入 R（万元）与电台广告费 x_1（万元）和报纸广告费 x_2（万元）间的关系是

$$R = 15 + 14x_1 + 32x_2 - x_1x_2 - 2x_1^2 - 8x_2^2$$

试求：(1) 在广告费不受限制情况下的最优广告策略；

(2) 在广告费限制 1.5（万元）时，其相应的最优广告策略。

解：$R(x_1, x_2) = 15 + 14x_1 + 32x_2 - x_1x_2 - 2x_1^2 - 8x_2^2$

(1) $\left.\begin{aligned} R'_{x_1} &= 14 - x_2 - 4x_1 = 0 \\ R'_{x_2} &= 32 - x_1 - 16x_2 = 0 \end{aligned}\right\} \Rightarrow$ 解得 $\begin{cases} x_1 = \frac{64}{21} \\ x_2 = \frac{38}{21} \end{cases}$ 此时 $R_{max} = R(\vec{x}) = \frac{457}{7} \doteq 65.3$

(2) s.t.: $x_1 + x_2 = 1.5$.

$L(x_1, x_2, \lambda) = 15 + 14x_1 + 32x_2 - x_1x_2 - 2x_1^2 - 8x_2^2 + \lambda(x_1 + x_2 - 1.5)$

$$\left.\begin{aligned} L'_{x_1} &= 14 - x_2 - 4x_1 + \lambda = 0 \\ L'_{x_2} &= 32 - x_1 - 16x_2 + \lambda = 0 \\ L'_{\lambda} &= x_1 + x_2 - 1.5 = 0. \end{aligned}\right\} \Rightarrow \begin{cases} x_1 = \frac{1}{4} \\ x_2 = \frac{5}{4} \end{cases}$$

此时 $R_{max} = R(\frac{1}{4}, \frac{5}{4}) = \frac{729}{16} \doteq 45.5$

P. 74. 编程计算实习题